

Ultrahigh Mass Activity Pt Entities Consisting of Pt Single atoms, Clusters, and Nanoparticles for Improved Hydrogen Evolution Reaction

Zhiqiang Zeng, Sven Küspert, S. Esmael Balaghi, Haytham E. M. Hussein, Niklas Ortlieb, Markus Knäbbeler-Buß, Philipp Hügenell, Stephan Pollitt, Niclas Hug, Julia Melke, and Anna Fischer*

Platinum is one of the best-performing catalysts for the hydrogen evolution reaction (HER). However, high cost and scarcity severely hinder the large-scale application of Pt electrocatalysts. Constructing highly dispersed ultrasmall Platinum entities is thereby a very effective strategy to increase Pt utilization and mass activities, and reduce costs. Herein, highly dispersed Pt entities composed of a mixture of Pt single atoms, clusters, and nanoparticles are synthesized on mesoporous N-doped carbon nanospheres. The presence of Pt single atoms, clusters, and nanoparticles is demonstrated by combining among others aberration-corrected annular dark-field scanning transmission electron microscopy, X-ray absorption spectroscopy, and electrochemical CO stripping. The best catalyst exhibits excellent geometric and Pt HER mass activity, respectively ≈ 4 and 26 times higher than that of a commercial Pt/C reference and a Pt catalyst supported on nonporous N-doped carbon nanofibers with similar Pt loadings. Noteworthy, after optimization of the geometrical Pt electrode loading, the best catalyst exhibits ultrahigh Pt and catalyst mass activities ($56 \pm 3 \text{ A mg}^{-1}_{\text{Pt}}$ and $11.7 \pm 0.6 \text{ A mg}^{-1}_{\text{Cat}}$ at $-50 \text{ mV vs. reversible hydrogen electrode}$), which are respectively ≈ 1.5 and 58 times higher than the highest Pt and catalyst mass activities for Pt single-atom and cluster-based catalysts reported so far.

1. Introduction

In the context of the mandatory decarbonization of the energy and chemistry sector, green hydrogen plays a key role as energy carrier (fuel) and platform molecule, and represents a sustainable alternative to fossil-based ones.^[1–3] Water electrolysis, i.e., the electrochemical process of H_2O splitting into H_2 and O_2 , is considered as one of the most promising technologies to produce high-purity H_2 from renewable energy sources, i.e., green hydrogen, and the maturity of this technology is proven by its industrial implementation.^[4] Among the known electrolyzer technologies including alkaline, proton exchange membrane, anion exchange membrane, or solid oxide water electrolysis, proton exchange membrane water electrolysis (PEMWE), based on water electrolysis in acidic conditions, is a very promising technology which is currently deployed, due to

Z. Zeng, S. Küspert, N. Ortlieb, N. Hug, J. Melke, A. Fischer
Institute of Inorganic and Analytical Chemistry (IAAC)
University of Freiburg
Albertstraße 21, 79104 Freiburg, Germany
E-mail: anna.fischer@ac.uni-freiburg.de

Z. Zeng, S. Küspert, S. E. Balaghi, J. Melke, A. Fischer
Freiburg Materials Research Center (FMR)
University of Freiburg
Stefan-Meier-Straße 21, 79104 Freiburg, Germany

S. E. Balaghi, N. Ortlieb, J. Melke, A. Fischer
Freiburg Center for Interactive Materials
and Bioinspired Technologies (FIT)
University of Freiburg

Georges-Köhler-Allee 105, 79110 Freiburg, Germany

H. E. M. Hussein
Department of Chemistry
University of Warwick
Coventry CV4 7AL, UK

M. Knäbbeler-Buß, P. Hügenell
The Fraunhofer Institute for Solar Energy Systems ISE
Heidenhofstraße 2, 79110 Freiburg, Germany

S. Pollitt
Paul Scherrer Institute
Forschungsstrasse 111, Villigen PSI, Villigen 5232, Switzerland

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/smll.202205885>.

© 2023 The Authors. Small published by Wiley-VCH GmbH. This is an open access article under the terms of the Creative Commons Attribution-NonCommercial License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

DOI: 10.1002/smll.202205885

several advantages including high operating current densities, high purity of gases, high outlet pressures, and a small footprint when compared to alkaline water electrolysis.^[4] However, the cost of PEMWE is still very high due to the high cost of the catalyst-coated membranes containing high amounts of platinum-group-metals (such as Pt, Ir, or Ru) accounting for 23%–47% of the manufacturing costs at the moment.^[5,6] Therefore, strategies need to be implemented to either reduce noble metal loadings and/or increase overall energy conversion efficiency by designing highly active and durable electrocatalysts. On the cathode side, Pt is still thought to be the best-performing electrocatalyst for the hydrogen evolution reaction (HER) from water.^[7,8] However, its high cost (almost US\$ 30 g^{−1})^[9] and scarcity (crustal abundance ≈0.005 ppm)^[10] severely hinder the large-scale application of Pt-based electrocatalysts. Therefore, Pt reduction on the cathode side with maintained high catalyst activity and stability is a major challenge in PEMWE development. Future Pt loadings of 0.05 mg cm^{−2} is proposed, representing a reduction in Pt of 20 to 40 times regarding current Pt loadings.^[6] Considering state-of-the-art Pt/C electrocatalysts consisting of supported Pt nanoparticles (NPs) on carbon, only surface and subsurface atoms are accessible to reactants during the electrocatalytic process. Thus, reducing the size of the Pt NPs is sought as an effective method to increase Pt atom utilization and reduce the cost of Pt-based electrocatalysts.^[7,11] In addition, increasing the number of catalytically active sites by optimizing the dispersion of Pt entities (such as Pt nanoparticles (NPs), Pt clusters, and Pt single atoms) without changing the Pt mass loading, promotes the whole electrode's activity.^[12–14] Therefore, developing synthesis methods for highly dispersed ultrasmall Pt-entity-based catalysts for HER is important to support the widespread increase in PEM electrolyzers installation capacities.^[15,16]

Dry and wet synthetic approaches have been widely used for fabricating Pt-based electrocatalysts.^[17–23] Although dry synthetic approaches, such as physical vapor deposition^[17] and atomic layer deposition methods,^[7,18] allow fabricating highly dispersed Pt ultrasmall nanoparticles,^[17,18] and even clusters and single atoms,^[7] these methods need expensive facilities. Moreover, not all Pt atoms are deposited on the support during synthesis, as a large part of them are either deposited on the inner reactor wall or pumped out.^[24–26]

Among various wet synthetic approaches, impregnation is the simplest and most widespread method for Pt catalyst preparation.^[27,28] Mesoporous support structures have been commonly used in this method because they can effectively inhibit metal (atoms/precursors) from diffusion and aggregation in nearby pores to form highly dispersed ultrasmall Pt NPs, and even clusters and single atoms. Mesoporous carbons are attractive as catalyst supports because of their large surface area, high conductivity, controlled mesoporous structure, and excellent stability in both acid and basic solutions.^[29,30] Furthermore, N-doping of mesoporous carbons can increase Pt dispersion by enabling strong Pt–N interactions between the N-functionalities of N-doped carbons and either Pt precursors^[31] or Pt NPs,^[32–34] (or Pt clusters and single sites) where the strong Pt–N interactions can also stabilize Pt and inhibit Pt entity detachment.^[31]

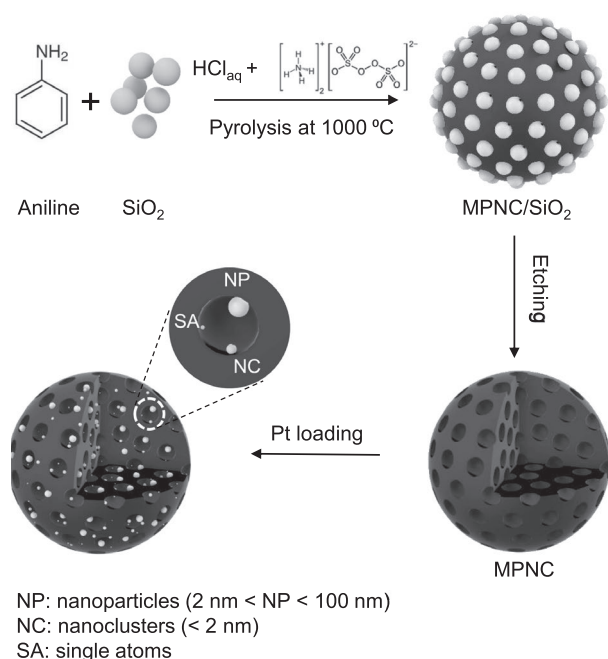
In this work, we synthesized highly dispersed Pt entities supported on mesoporous N-doped carbon (MPNC) nanospheres

for improved HER. The MPNC nanospheres, derived from the pyrolysis of polyaniline–silica nanocomposites, are highly monodisperse with a particle size around 298 ± 23 nm and highly porous with interconnected mesopores of 24 ± 2 nm throughout the entire particle volume. By incipient wetness impregnation, 20 wt% Pt was deposited onto these polyaniline-derived MPNC nanospheres as well as on nonporous polyaniline-derived N-doped carbon nanofibers (CNF) as supports, denoted as Pt/MPNC and Pt/CNF, respectively. By comparing MPNC nanospheres and CNF as supports, we found that the mesoporous structure of the MPNC nanospheres was favorable for the formation of highly dispersed Pt single atoms, clusters and NPs, as demonstrated by a combination of X-ray diffraction analysis (XRD), X-ray photoelectron spectroscopy (XPS), aberration-corrected annular dark-field scanning transmission electron microscopy (ac-ADF-STEM), X-ray absorption spectroscopy (XAS), and electrochemical CO stripping. The highly dispersed Pt entities endowed the Pt/MPNC catalyst with a large electrochemical surface area (ECSA) and an excellent HER geometric and Pt mass activity. By comparing the different samples featuring different Pt speciations, we could estimate that the Pt single atoms account for around 65% of the measured absolute HER current, while the Pt NP and clusters account for the remaining 35%. This excellent HER activity turned out to be respectively 4.2 and 26 times higher than that of a commercial Pt/C reference and a homemade Pt/CNF catalyst with similar Pt loading (20 wt% Pt). Noteworthy, after optimization of the geometrical Pt electrode loading, the best catalyst exhibits ultrahigh Pt and catalyst mass activities (56 ± 3 A mg^{−1}_{Pt} and 11.7 ± 0.6 A mg^{−1}_{Cat} at −50 mV vs. reversible hydrogen electrode (RHE)), which are respectively 1.5 ± 0.1 and 58 ± 3 times higher than the highest Pt and catalyst mass activities (determined at the same overpotential and Pt electrode loading) of Pt single-atom and cluster-based catalysts reported so far.^[35–39]

2. Results and Discussions

2.1. Synthetic Approach

In electrocatalysis, understanding structure-property-activity relationships forms one of the basis to improve the kinetics of the physicochemical processes that underpin the HER. We herein design our experiment to demonstrate the structural advantages of the MPNC nanospheres and their mesoporous structure. First, polyaniline-derived MPNC nanospheres were synthesized according to **Scheme 1**, where aniline (as N-containing carbon source) and SiO₂ nanospheres (as template) were homogenized in an acidic solution. After the addition of ammonium persulfate and upon completion of the reaction, followed by washing and drying, a green powder composed of polyaniline–SiO₂ (PANI–SiO₂) composite nanoparticles was collected. The solid phase was pyrolyzed under N₂ at a carbonization temperature of 1000 °C.^[29,30] Note that CNF (N-doped carbon nanofibers) were made using similar synthesis parameters, i.e., carbonization of a polyaniline structure, but omitting the SiO₂ template. Upon obtaining the support materials MPNC and CNF, we incorporated Pt by incipient wetness impregnation and subsequent reduction at 300 °C under 10 vol% of H₂/Ar gas for 2 h, yielding



Scheme 1. Schematic illustration of the synthesis of highly dispersed Pt entities composed of a mixture of Pt single atoms, clusters, and nanoparticles incorporated into MPNC nanospheres. The high Pt-dispersion achieved results from the mesopore-confinement effects of the support.

Pt-based electrocatalysts denoted hereafter as Pt/MPNC and Pt/CNF. Note that for all experiments, the support materials were loaded with 20 wt% Pt unless otherwise stated.

2.2. Structural Characterization of Support Materials and Electrocatalysts

The morphology and structure of MPNC and CNF carbon supports were characterized by scanning electron microscopy (SEM), as shown in **Figure 1A–C**. The monodisperse MPNC nanospheres with a particle size of 298 ± 23 nm are highly porous with spherical mesopores running throughout the entire particles, in which the average mesopore size was determined by high resolution SEM (**Figure 1A,B**) and nitrogen physisorption (**Figure 1D–E**) to be 24 ± 2 nm (visible pores at the outer surface of the spheres) and 28 nm, respectively. As a result, the MPNC nanospheres exhibit a large specific surface area (SSA) of $540 \text{ m}^2 \text{ g}^{-1}$, determined by the Brunauer-Emmett-Teller (BET) method, which is approximately 10 times larger than the BET SSA of the CNF, revealing a nonporous nanofiber structure (**Figure 1C**) with a low SSA of $52 \text{ m}^2 \text{ g}^{-1}$ (BET method; **Figure 1D** and **Table 1**). In addition to mesopores, the MPNC nanospheres also contain smaller micropores (i.e., pores < 2 nm) as shown in **Figure 1E**. The pore size distribution (determined by quenched solid density functional theory (QSDFT) analysis using a spherical, cylindrical, and slit pore model) reveals the presence of both mesopores and micropores in the MPNC nanospheres (**Figure 1E**), in which the micropores and the mesopores respectively account for 42.4% and 57.6% of the SSA and for 7.2% and 92.8% of the specific pore volume.

In addition to examining the morphology and porosity characteristics of the MPNC and CNF supports, we also determined their elemental composition and graphitization. The MPNC and the CNF support materials show similar elemental compositions despite having different nanostructures. Both supports are composed of C, H, N, and S (**Figure 1F**; **Table S1**, Supporting Information), as revealed by combustion elemental analysis. The residuals are attributed to oxygen functionalities. The MPNC nanospheres have a slightly higher N content (4.11 wt%) than the CNF nanofibers (3.17 wt%). Since we anticipate that the graphitization of the MPNC and the CNF supports will impact their electronic conductivity,^[40] the graphitization and defect densities of both samples were investigated by carrying out Raman spectroscopy. The Raman spectra of both samples shown in **Figure 1G,H** exhibit two broad absorption bands. The first one approximately at 1352 cm^{-1} corresponds to the disorder-induced D band. The second one at 1600 cm^{-1} corresponds to the in-plane vibrational G band arising from the E_{2g} symmetry mode of the ideal graphitic lattice,^[41,42] in which the E_{2g} symmetry corresponds to the in-plane stretching vibration of sp^2 carbon. The D band can be deconvoluted into four D_i band contributions with $i = 1, 2, 3$, and 4. The $D1$ corresponds to a graphitic lattice vibration mode with A_{1g} symmetry (in-plane breathing vibrations of the aromatic ring structures), the $D2$ corresponds to a graphitic lattice mode with E_{2g} symmetry, the $D3$ originates from the amorphous carbon, and the $D4$ is assigned to the A_{1g} symmetry mode of the disordered graphitic lattice.^[43] The peak areas of the deconvoluted Raman bands of the two carbon supports were further compared, as shown in **Figure 1I**. Each corresponding peak of the MPNC and the CNF supports shows almost the same area, indicating that the two carbon supports have the same contribution in terms of amorphous phase and defects. The ratio of the D and G band intensities (I_D/I_G) is also used as indicative of the graphitization degree of carbon-based materials in literature.^[41,42] From **Figure 1G,H**, we found out that the graphitization degrees for MPNC and CNF are $I_D/I_G = 0.95$ and $I_D/I_G = 0.97$, respectively. The values of I_D/I_G ratio are almost similar, which evidently shows that the two carbon supports namely MPNC and CNF have similar graphitic structure/ordering. Consequently, we suggest that the major difference between the MPNC and the CNF supports is related to their nanostructure/porosity.

The two N-doped carbon supports, MPNC nanospheres and CNF, were loaded with 20 wt% of Pt by wetness impregnation and subsequent gas phase reduction (see the Experimental Section), resulting in Pt/MPNC and Pt/CNF catalysts. We benchmarked the lab-developed electrocatalysts against a commercial Pt/C reference catalyst (HiSPEC3000, AlfaAesar, 20 wt%) in terms of morphology, size, and structure, which we investigated using SEM, transmission electron microscopy (TEM), XRD, and ac-ADF-STEM, respectively.

First, we characterized the Pt crystallite structure of these three catalysts by XRD as shown in **Figure 2A,B**. Each sample shows five Pt diffraction reflexes, i.e., Pt (111), Pt (200), Pt (220), Pt (331), and Pt (222). In general, one can say that the sharper the reflexes, the larger the crystallite sizes.^[44] The Pt crystallite sizes were obtained by Rietveld refinement of the XRD patterns (see **Figure 2B**). Pt/MPNC shows the smallest Pt crystallite size of 2.2 nm, followed by Pt/C with 2.4 nm. Pt/CNF reveals

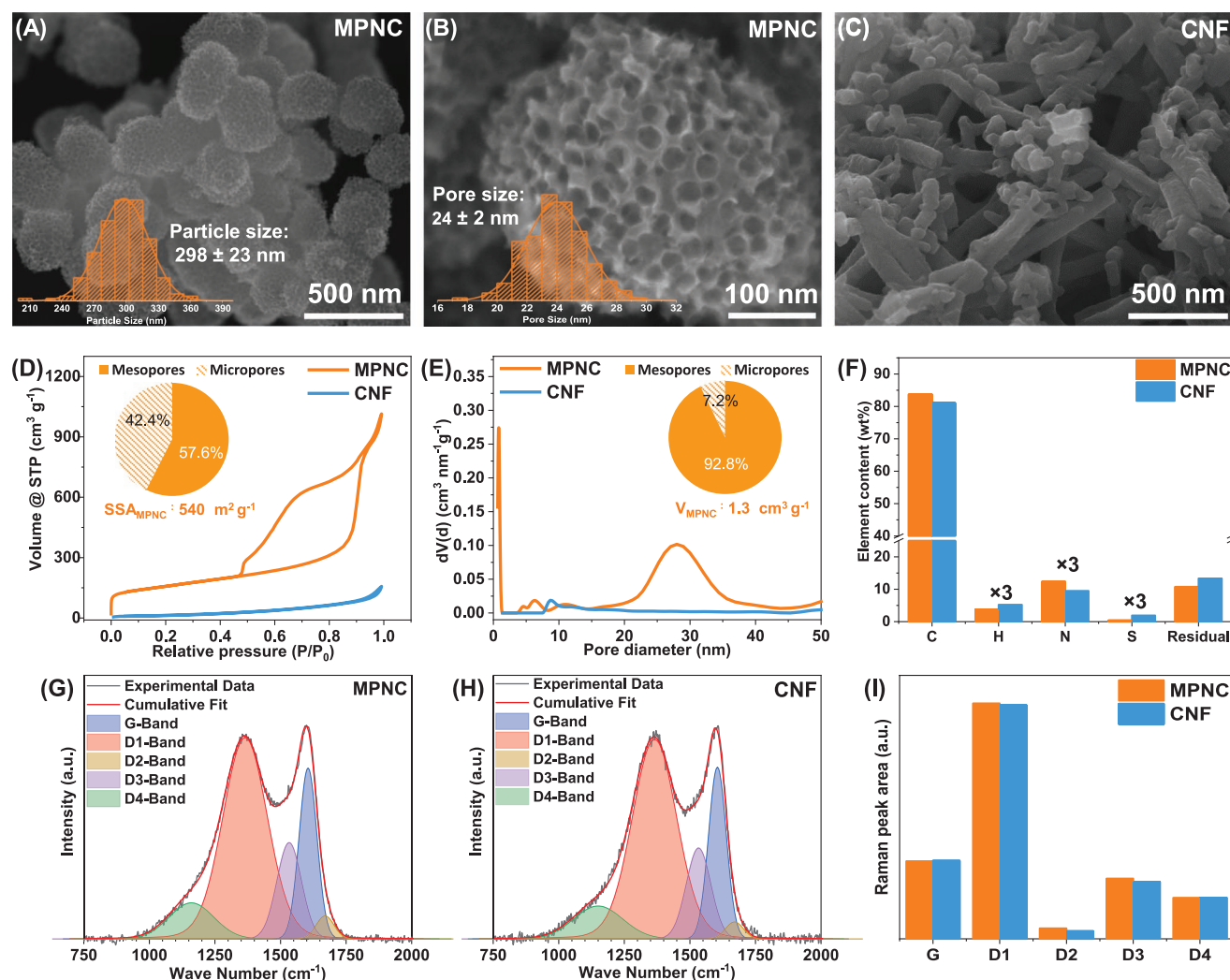


Figure 1. Characterizations of MPNC and CNF N-doped carbon supports. SEM images of A,B) MPNC, with the inset showing the corresponding particle and pore size distribution, and C) CNF. N₂ physisorption isotherms of D) MPNC (orange) and CNF (blue) and E) derived pore size distributions based on the QSDFT model taking into account spherical, cylindrical, and slit pores; the insets of (D) and (E) represent the mesopore and micropore contribution of MPNC to the specific surface area and pore volume, respectively. F) Bar plot of the elemental compositions (combustion analysis) showing the elements C, H, N, and S that make up the MPNC and CNF supports. The residuals are attributed to oxygen. G,H) Raman spectra of MPNC and CNF, respectively. I) bar plot showing the peak areas of the deconvoluted Raman bands for MPNC (orange) and CNF (blue).

two larger Pt crystallite sizes of 4 and 14 nm, which account respectively for 16 and 84 at% of the immobilized crystalline Pt species visible in XRD.

In addition, the Pt/MPNC, Pt/CNF and Pt/C catalysts were characterized using SEM and TEM as shown in Figure 2C–E and Figure S4 (Supporting Information), in which Pt/MPNC exhibits the smallest Pt NP size of 1.5 ± 0.6 nm compared with Pt/CNF (16.3 ± 4.4 nm) and Pt/C (2.9 ± 0.6 nm). This is attributed to the structural/morphological advantage of the MPNC support, i.e., its continuous mesoporous structure, which significantly reduces aggregation of Pt atoms, favoring the formation of highly dispersed Pt entities. In contrast, the non-porous structure of the CNF support allows Pt atom diffusion and aggregation into larger Pt NPs during the incipient wetness impregnation and reduction step. The slight size differences between particle and crystallite sizes determined by TEM (number weighted particle size) and XRD Rietveld refinement

(volume weighted particle size) may stem from the used analysis method (type of distribution), the extension of the analysis method, i.e., local (only a few spots of catalyst particles in TEM) vs. global (a full catalyst powder in XRD) analysis, as well as the chemical nature of the Pt entities, i.e., some Pt potentially being amorphous Pt or Pt oxide, which cannot be detected by XRD.^[45] Nevertheless, based on the above analysis, we could confirm that the MPNC support has a clear structural/morphological advantage, i.e., its mesopore-confinement effect, inhibiting Pt atoms from diffusion and aggregation during incipient wetness impregnation and reduction, in other words during catalyst synthesis. Therefore, the porous structure/morphology of MPNC nanospheres favors the formation of highly dispersed ultrasmall Pt entities.

To evaluate if besides the Pt NPs revealed by TEM and XRD, smaller Pt entities such as clusters and single atoms may be present at the surface of the three different catalysts, ac-ADF-STEM

Table 1. Comparison of the different catalysts studied in this paper in terms of BET specific surface area (SSA), Pt nanoparticle (NP) diameter and Pt nanocrystallite size, electrochemical surface area (ECSA), geometric current density, and Pt mass activity, in which Pt/MPNC, Pt/CNF, and Pt/C have the same Pt loading on the support (20 wt% Pt) and on the electrode ($12.2 \mu\text{g}_{\text{Pt}} \text{cm}^{-2}_{\text{geo}}$) during the electrochemical test.

Sample	MPNC	CNF	Pt/MPNC	Pt/CNF	Pt/C
BET SSA [$\text{m}^2 \text{g}^{-1}$]	540	52	412	53	219
Pt NP diameter TEM ^{a)} [nm]			1.5 ± 0.6	16.3 ± 4.4	2.9 ± 0.6
Pt crystallite size XRD ^{b)} [nm]			2.2	Phase 1: 4 Phase 2: 14	2.4
ECSA _H [$\text{m}^2 \text{g}^{-1}_{\text{Pt}}$] 0.5 M $\text{H}_2\text{SO}_4^{\text{c)}$			80 ± 6.9	4.4 ± 1.4	53 ± 1.2
ECSA _H [$\text{m}^2 \text{g}^{-1}_{\text{Pt}}$] 0.1 M $\text{HClO}_4^{\text{c)}$			83 ± 5.9	5.4 ± 1.1	55 ± 2.9
ECSA _{CO} [$\text{m}^2 \text{g}^{-1}_{\text{Pt}}$] 0.1 M $\text{HClO}_4^{\text{d)}$			136 ± 5.3	7.8 ± 0.8	73.4 ± 4.6
$Q_{\text{H}}/Q_{\text{CO}}^{\text{e)}$			0.31 ± 0.03	0.34 ± 0.03	0.38 ± 0.01
ECSA _H /ECSA _{CO} ^{e)}			0.61 ± 0.07	0.69 ± 0.07	0.75 ± 0.02
Pt mass activity [$\text{A mg}^{-1}_{\text{Pt}}$] ^{f)}			5.45 ± 0.35	0.21 ± 0.08	1.28 ± 0.16
Geometric current density [$\text{mA cm}^{-2}_{\text{geo}}$] ^{f)}	0.9 ± 0.04	0.1 ± 0.02	68 ± 4.2	2.6 ± 0.9	16 ± 2

^{a)}Number-weighted particle diameter with standard deviation based on the size quantification of 200 particles; ^{b)}Based on Rietveld refinement of XRD pattern; ^{c)}Based on hydrogen adsorption peak; ^{d)}Based on electrochemical CO stripping; ^{e)}Obtained in 0.1 M HClO_4 ; ^{f)}At an overpotential of 20 mV vs. RHE

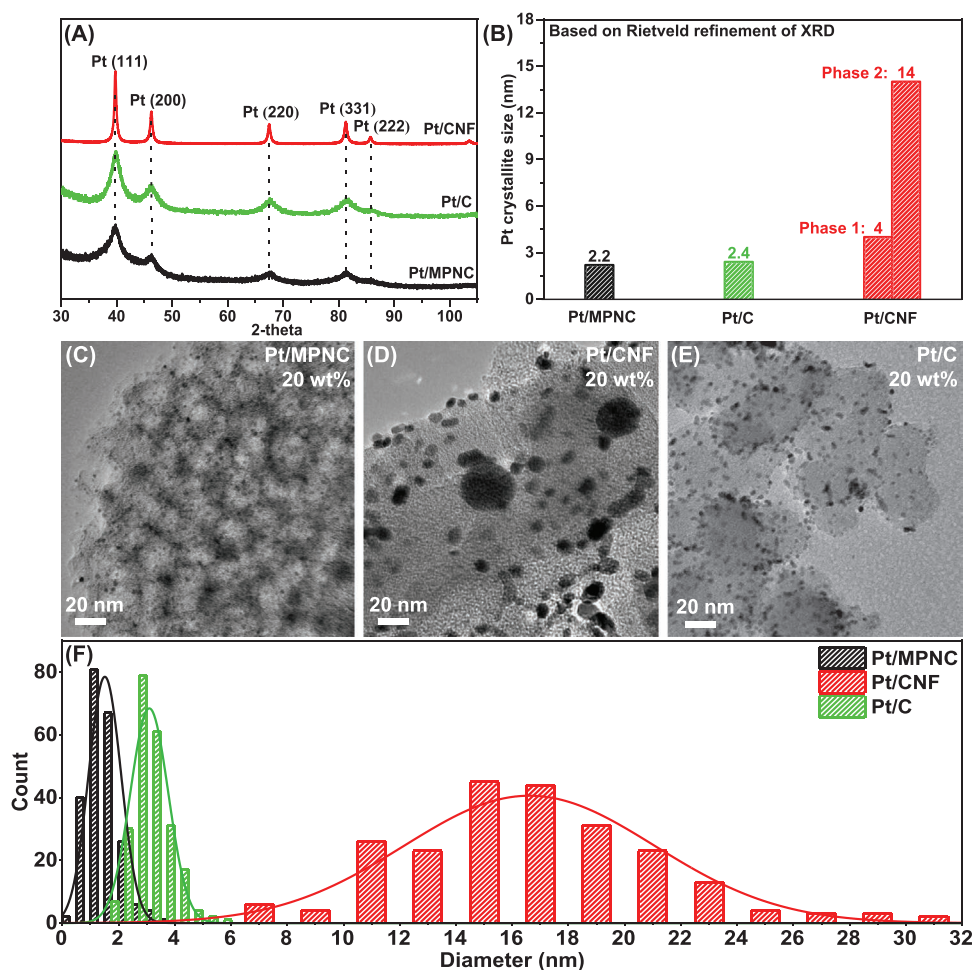


Figure 2. Pt nanoparticle characterization in the resulting catalysts. XRD patterns A) and corresponding Pt crystallite sizes B) of Pt/MPNC, Pt/CNF and Pt/C catalysts. TEM images of C) Pt/MPNC, D) Pt/CNF, and E) a commercial 20 wt% Pt/C (HiSPEC3000) catalyst as well as F) corresponding Pt nanoparticle size distributions.

measurements were performed. As can be seen in **Figure 3** and Figures S7–S9 (Supporting Information), the atomic resolution ac-ADF-STEM images reveal that 1) the MPNC support surface in the Pt/MPNC catalyst is decorated with a high density of Pt single atoms, atom clusters, and tiny Pt NPs (≈ 2 nm), see Figure 3A–F; 2) the CNF support surface in the Pt/CNF catalyst is decorated with a low density of Pt single atoms and larger Pt NPs, see Figure 3G–L, and 3) the commercial Pt/C

catalyst is dominated by Pt NPs without observing any Pt single atoms suggesting the complete absence of Pt single atoms in the commercial sample (see Figure S9 in the Supporting Information). Furthermore, we adapted STEM imaging quantification (see SI1 for the associated method and Figure S10 in the Supporting Information) to provide a visualization of the atomic columns that build up the various Pt entities found on these three catalysts, i.e., Pt/MPNC, Pt/CNF, and Pt/C. The

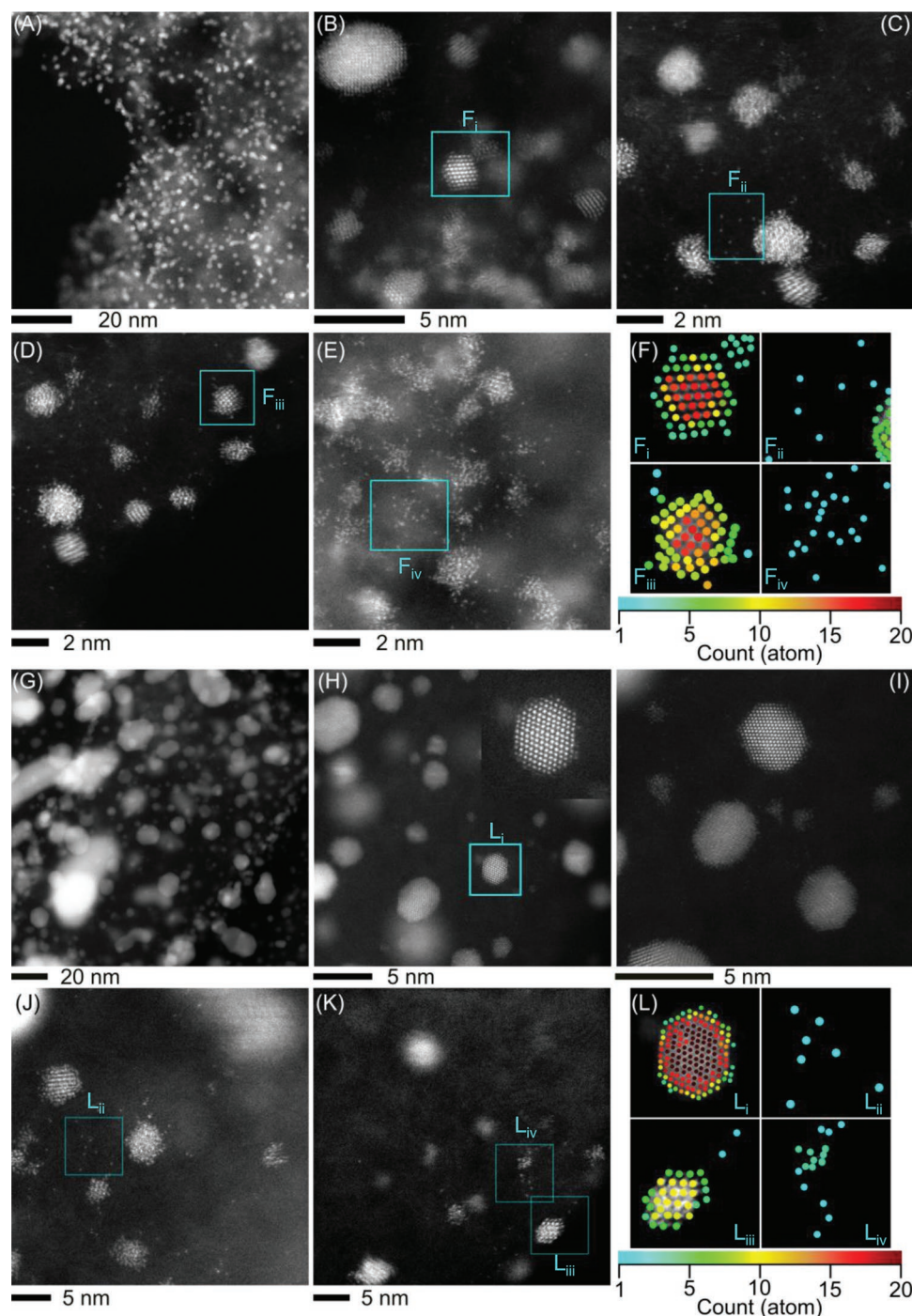


Figure 3. A–E) ac-ADF-STEM images of Pt/MPNC and F) provides quantification and demonstration of the presence of high density of Pt single atoms. G–K) ac-ADF-STEM images of Pt/CNF and L) provides information on the presence of low density of Pt single atoms.

image quantification is demonstrated by color coded ac-ADF-STEM images in Figure 3F for Pt/MPNC and 3L for Pt/CNF, while Figure S9C (Supporting Information) provides the ones for Pt/C. Single atom columns are represented as aqua color dots, and it is possible to see that Pt/MPNC catalysts feature a high density of Pt single atoms in comparison to the Pt/CNF catalyst. On the other hand, the Pt/C catalyst is free of Pt single atoms. We additionally evidenced the presence of single sites by analyzing the intensity distribution along selected cross-lines cutting through high-contrast dots, presumably Pt single atoms, frequently present in the ac-ADF-STEM images of the Pt/MPNC sample (see Figure S8 in the Supporting Information). Distinct intensity peaks with a base peak width of around 0.1–0.2 nm are revealed (Figure S8C, Supporting Information), which fall in the range of the size of a Pt single atoms.^[46] Such observations suggest that both the mesopore-confinement effect as well as the N-doping characteristics of the MPNC lead to favoring the formation of highly dispersed Pt entities. In addition, this demonstrates that the characterization of Pt-based catalysts solely by XRD and low-resolution TEM, as is often done in literature, is not sufficient to get a full picture of the complete Pt entities of a catalyst. This is even more true in the case of N-doped carbon supports, as N-surface functionalities are known to stabilize highly dispersed Pt entities such as Pt clusters and Pt single sites;^[36,47] the latter being invisible in regular TEM and XRD measurements.

The various Pt entities revealed by the described combinatorial analysis above, involving XRD, TEM, and ac-ADF-STEM, were further probed by electrochemical CO stripping experiments, as shown in Figure 4A. The observed shape or peak multiplicity of CO oxidation via stripping experiments can have various causes which can be attributed to, e.g., surface defects, support effects, particle size, particle size distribution, faceting, or applied experimental protocol as shown in the review work of Ciapina et al.^[48] It can be assumed that particle size, also directing crystal faceting, has the greatest influence on the peak position of CO oxidation since they directly affect the adsorption energy of CO_{ads} and the required OH_{ads}. Therefore, in the following, peak position is used as an indicator for particle size and compared with the results from XRD, TEM, and ac-ADF-STEM. All CO stripping experiments were conducted in 0.1 M HClO₄ solution to decrease the effect of anion adsorption on the Pt surface, which is the case for the strong adsorbing sulfate anion.^[48,49] As can be seen in Figure 4A, Pt/CNF shows indeed two CO oxidation peaks, the first one (Peak 1) at 725 mV vs. RHE is related to either agglomerated Pt NPs or large Pt NPs on the CNF support. The presence of such Pt entities is proven by XRD and TEM as shown in Figure 2. However, a proper disentanglement of a size or an agglomeration effect, which leads to CO stripping oxidation at lower potentials, is still under discussion in literature and out of the scope of this work.^[48] The second peak (Peak 2) at 810 mV vs. RHE is attributed to isolated Pt NPs.^[50] Pt/C shows a small agglomeration-related shoulder at around 725 mV vs. RHE. In this case, the attribution to agglomeration is more rational as no larger Pt NPs are found by XRD and TEM analysis. Furthermore, a sharp peak at 810 mV vs. RHE (Peak 2), indicates that Pt/C is mainly composed of isolated Pt NPs and that Pt NPs do not aggregate seriously as in the case of Pt/CNF. Pt/MPNC reveals two CO

oxidation peaks. Among them, a sharp peak describing small nonaggregated Pt NPs is found at 810 mV vs. RHE (Peak 2), while in contrast to Pt/C and Pt/CNF a second broad one (Peak 3) is located at around 886 mV vs. RHE. Very importantly, when compared to the other samples, this peak 3 is ascribed to the existence of sub-nanometer Pt NPs, i.e., Pt atom clusters, on the MPNC support.^[51] The different Pt entities on MPNC, namely Pt NPs and clusters found by CO stripping, are consistent with the ac-ADF-STEM analysis.

By integrating the charge transferred during CO oxidation (Q_{CO}) and normalizing it by the commonly used value for the two-electron CO oxidation process on a polycrystalline Pt surface of $420 \mu\text{C cm}^{-2}_{Pt}$ (as a twofold of $210 \mu\text{C cm}^{-2}_{Pt}$, the value used for the one-electron hydrogen adsorption process on a polycrystalline Pt surface with mainly (100), (110), and (111) planes^[52]) and the used Pt mass, the ECSA_{CO} was calculated (Table 1). Correspondingly, the ECSA_H was also determined based on the one-electron process of hydrogen underpotential deposition by integration of the hydrogen adsorption charge Q_H in the cyclic voltammograms (Figure S17A, Supporting Information; Table 1). While comparing the individual ECSA values, it is noticeable that ECSA_{CO} is higher than ECSA_H for all catalysts (ECSA_H vs. ECSA_{CO}: 83 ± 5.9 vs. $136 \pm 5.3 \text{ m}^2 \text{ g}^{-1}_{Pt}$ for Pt/MPNC; 5.4 ± 1.1 vs. $7.8 \pm 0.8 \text{ m}^2 \text{ g}^{-1}_{Pt}$ for Pt/CNF; and 55 ± 2.9 vs. $73.4 \pm 4.6 \text{ m}^2 \text{ g}^{-1}_{Pt}$ for Pt/C), indicating $Q_H/Q_{CO} < 0.5$ in all cases as shown in Table 1. The ECSA_H/ECSA_{CO} ratios of Pt/MPNC, Pt/CNF and Pt/C catalysts were also compared in Figure 4B and Table 1, and are respectively 0.61 ± 0.02 , 0.69 ± 0.07 , and 0.75 ± 0.02 . Noteworthy, Pt catalysts with N-doped carbon supports exhibited a lower ECSA_H/ECSA_{CO} ratio. This discrepancy is assigned to multiple factors which are belonging to the nature of Pt crystallite size, agglomeration and defects controlling the CO and hydroxide ad/desorption as well as their mobility on the Pt surface.^[53–55] Maillard et al. also found a shift of the ideal charge transfer ratio $Q_H/Q_{CO} = 0.5$ to lower values of ≈ 0.4 for Pt particles $\leq 2 \text{ nm}$.^[56] In addition to the above-mentioned hypotheses, they pointed out that adsorption of multiple CO molecules per active Pt site is observed for Pt carbonyl clusters. In the case of subnanometer Pt clusters, Huang et al. have shown a high ECSA_{CO} up to $300 \text{ m}^2 \text{ g}^{-1}_{Pt}$ for 0.8 nm sized Pt clusters, but could not calculate ECSA_H as no defined H_{upd} features were detected by cyclic voltammetry (CV).^[51] So taking different theoretical approaches and experimental results into account, a full explanation is yet not found to the best of our knowledge, but the CO-oxidation Peak 3 strongly points to the presence of Pt atom clusters, in line with our ac-ADF-STEM results.

In addition to the mesopore-confinement effect, N-doping of the MPNC support plays a decisive role in the formation of highly dispersed Pt entities due to strong interactions between Pt and the N functionalities.^[34,57,58] Therefore, the N contents and distributions of Pt/MPNC and Pt/CNF were studied systematically by elemental analysis and STEM X-ray energy dispersive spectroscopy (EDS) mapping. The elemental composition in Figure S16A in the Supporting Information demonstrates that Pt/MPNC and Pt/CNF have almost the same N content (3.18 wt% vs 2.85 wt%, respectively). The STEM-EDS mappings of Pt/MPNC and Pt/CNF in Figure S5A,B (Supporting Information) confirm that C, N, and O are homogeneously

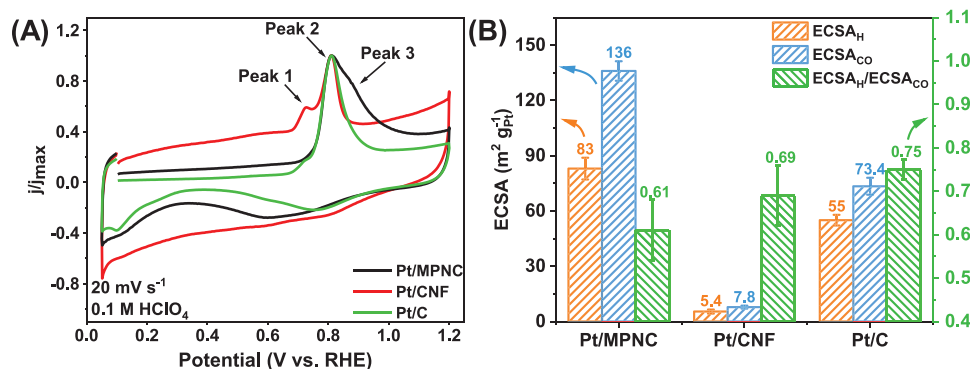


Figure 4. A) CO-stripping voltammograms normalized by the maximum current density (i/j_{max}) and B) ECSA comparison of Pt/MPNC, Pt/CNF, and Pt/C catalysts.

distributed throughout the material, but Pt is only distributed homogeneously on the MPNC support, whereas Pt on the CNF support exhibits aggregation.

XPS analysis was further used to examine the N-doping and valence state of Pt species for Pt/MPNC and Pt/CNF as well as Pt/C. The high-resolution N 1s XPS spectra in Figure 5A,B show five different bonding configurations of N atoms. Besides the four common types of N found in N-doped carbons,^[58–60] pyridinic N (398.1 eV), pyrrolic N (400.8 eV), graphitic N (401.3 eV), and oxidized N (405.1 eV), Pt–N bonding (399.1 eV) was also observed on Pt/MPNC and Pt/CNF,^[60] verifying the presence of both N-doping and of a strong interaction between Pt atoms and N atoms on Pt/MPNC and Pt/CNF. Noteworthy, Pt/MPNC has a higher Pt–N percentage (13%) than Pt/CNF (4%) due to the presence of the higher density and smaller Pt entities (clusters and single sites) on MPNC where more electron transfer from Pt atoms and N atoms is possible, resulting in a negative binding energy shift for the four common N-species of Pt/MPNC as shown in Figure 5A.^[61,62] In contrast, no binding energy shift was observed in Figure 5B for the four N-species of Pt/CNF, probably due to the lower density of Pt–N sites and bigger Pt entities on CNF, and no N 1s peak was observed on Pt/C in Figure S14C (Supporting Information), verifying the absence of N-doping on the Pt/C reference catalyst, in agreement with the previous elemental analysis results (Figure S16A and Table S1, Supporting Information). The Pt 4f spectra of these three catalysts exhibit two peaks as shown in Figure 5C,D and Figure S15 (Supporting Information), corresponding to Pt 4f_{7/2} and Pt 4f_{5/2} states due to the spin–orbit coupling. The two peaks can be fitted into three standard valence states, namely, Pt⁰, Pt²⁺, and Pt⁴⁺. The fit results are shown in Table S3 (Supporting Information), and for better visibility, background-corrected Pt 4f spectra are shown in Figure 5C,D. Pt single atoms were reportedly mainly in the Pt²⁺ species,^[58] therefore, the peak area ratios of Pt²⁺/Pt⁴⁺ would increase when Pt single atoms exist on the supports. Pt/MPNC and Pt/CNF in Figure 5E show an increased Pt²⁺/Pt⁴⁺ ratio compared with Pt/C, pointing to the existence of Pt single atoms in Pt/MPNC and Pt/CNF. Thus, N-doping can anchor and stabilize Pt single atoms on N-doped carbon supports. In addition to the highest Pt²⁺/Pt⁴⁺ ratio, Pt/MPNC also has the highest percentage of Pt oxide phases among these three catalysts shown in Figure 5E owing to a size effect,^[63] i.e., the percentage of surface oxidized Pt increases as Pt particle size decreases.

In addition, the chemical state and an overall picture of the coordination environment of the Pt atoms in the bulk of the samples were further investigated by X-ray absorption spectroscopy (XAS). The results for Pt/MPNC, along with the reference samples (Pt/CNF, Pt/C, Pt foil, and PtO)

2), are shown in Figure 6. As shown in Figure 6A, the Pt L₃-edge white line intensity in the X-ray absorption near-edge structure (XANES) region has been compared with the reference materials. The Pt L₃-edge white line intensity is related to the electronic transitions from 2p_{2/3} to an unoccupied 5d state and a higher white line intensity indicates a higher average oxidation state for the Pt centers.^[64,65] The white line intensity for Pt/MPNC was higher than for the Pt/C and Pt/CNF, indicating a higher oxidation state for Pt/MPNC. In contrast, the Pt/C and Pt/CNF show a similar white line intensity to the one of the Pt foil, indicating more Pt⁰ metallic character in their structure and larger Pt NPs.^[65,66] This is in agreement with the different Pt entities existing in Pt/MPNC, Pt/CNF and Pt/C, as revealed by XRD, TEM, ac-ADF-STEM, and electrochemical CO stripping analysis. Indeed, the lower surface-to-volume ratio of the larger Pt particles and crystallites existing in Pt/CNF, results in the presence of less Pt surface oxides on these Pt particles (Pt NP size TEM: 16.3 ± 4.4 nm; XRD crystallite size: 4 and 14 nm) and hence contributes to a more metallic character, when compared to Pt/MPNC. The high oxidized state of Pt/MPNC, as shown by the XANES white line intensity, is due to the smaller size of the Pt NPs and clusters in Pt/MPNC (Pt NP size TEM: 1.5 ± 0.6 nm; XRD crystallite size: 2.2 nm), resulting in a larger surface-to-volume ratio and a higher amount of surface oxides, as well as the contributions of oxidized Pt single atoms in the N-doped carbon matrix. For Pt/C, mainly NPs in the size of 2.9 ± 0.6 nm and 2.4 nm are found by TEM and XRD, respectively, which is why the white line intensity is located between the one of Pt/CNF and Pt/MPNC. The formation of Pt–N bonds in Pt/MPNC was studied by Fourier transform (FT) k³-weighted extended X-ray absorption fine structure (EXAFS) (Figure 6B). Comparing FT-EXAFS spectra of Pt/MPNC, Pt/C, Pt/CNF, Pt foil, and PtO₂ (Figure 6B) revealed the dominant coordination shells around ≈1.6 Å which can be related to the overall contributions of Pt–N or Pt–O coordination shells around the Pt center.^[67] As shown in Figure 6C, the best-fitting results demonstrate that the dominant peak around 1.6 Å is ascribed to the overlap of Pt–N (of Pt single-atom structures) and Pt–O (of β-PtO₂ phase)

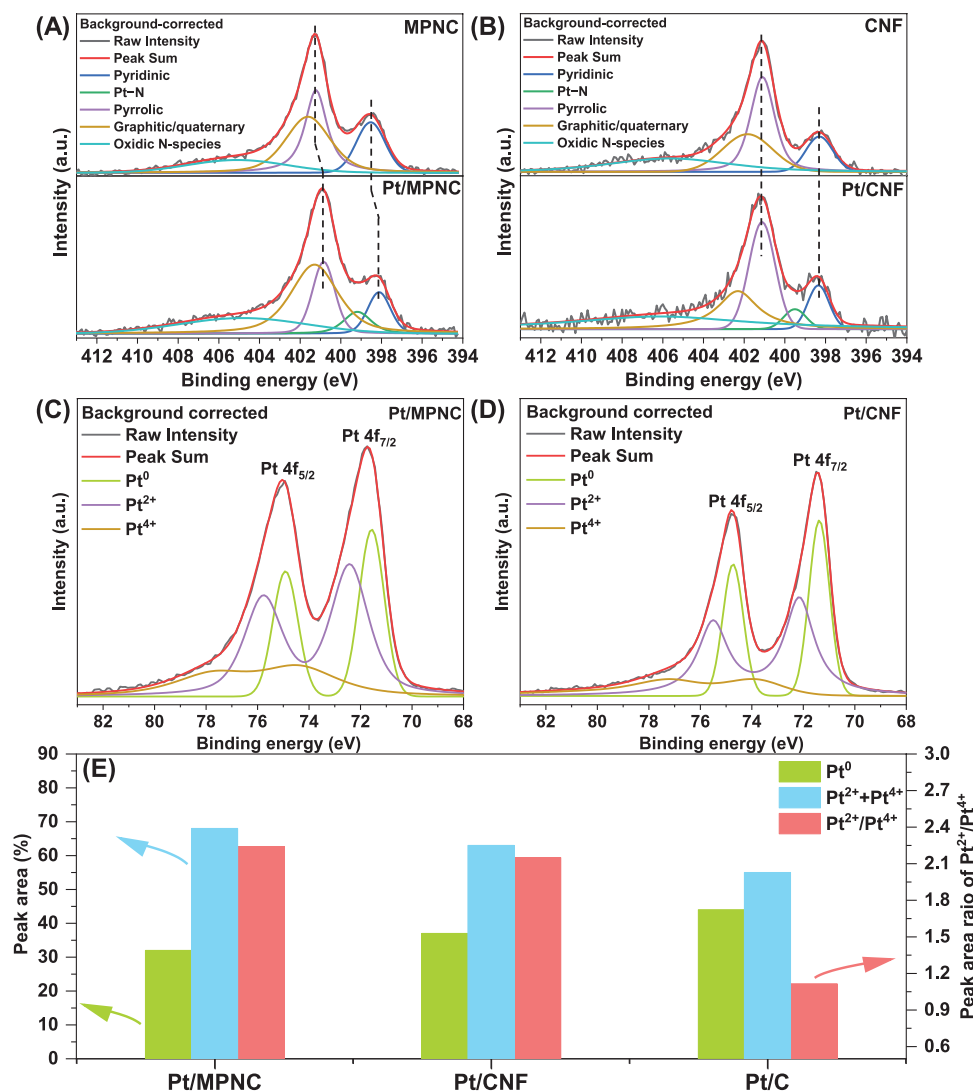


Figure 5. Surface characterization of the catalyst materials. N 1s XPS spectra of A) MPNC and Pt/MPNC and B) CNF and Pt/CNF. Pt 4f core level XPS spectra of C) Pt/MPNC and D) Pt/CNF. E) Peak area ratio of the single Pt 4f XPS components providing a comparison between the Pt/MPNC, Pt/CNF, and Pt/C catalysts based on the electronic structure.

first coordination shells (Figure S11 and Table S2, Supporting Information). Additionally, two minor peaks between $\approx 2\text{--}3\text{ \AA}$ are mainly ascribed to the contribution of the Pt–Pt coordination shell of the metallic Pt phase (Figure 6C; Table S2, Supporting Information). To further investigate the nature of the Pt/MPNC sample, Wavelet transform (WT) analysis of the Pt L₃-edge EXAFS oscillations was performed (Figure S12, Supporting Information). The conventional FT-EXAFS simulation cannot discriminate an unknown or complex system with overlapped scattering pathways in R space.^[68,69] In such circumstances, the WT technique can precisely help to resolve the coordination shells.^[70] Despite $\beta\text{-PtO}_2$, and Pt metallic foil, the WT-EXAFS of Pt/MPNC, showed only one major intensity maximum at $k \approx 5.8\text{ \AA}^{-1}$, which mainly represented the Pt–N and Pt–O contributions.

Though a full quantitative disentanglement between contributions of single atoms, clusters and NPs cannot be made on

the basis of XAS analysis, our correlative analytical approach of EXAFS, XRD, XPS and ac-ADF-STEM points without a doubt to the existence of Pt single atoms in Pt/MPNC. In more detail, the fitting contributions of Pt–N interactions by EXAFS and XPS analyses, which can only be detected if there is a sufficient amount of Pt single atoms present in the sample, as well as the microareal visibility of Pt single atoms by ac-ADF-STEM emphasize without a doubt the presence of Pt single atoms in Pt/MPNC.

2.3. Electrochemical Investigations—Activity and Stability for HER in Acidic Media

To demonstrate the advantages of the mesoporous structure and the related higher Pt dispersion in water electrolysis, we evaluated the electrocatalytic activities for the HER of the

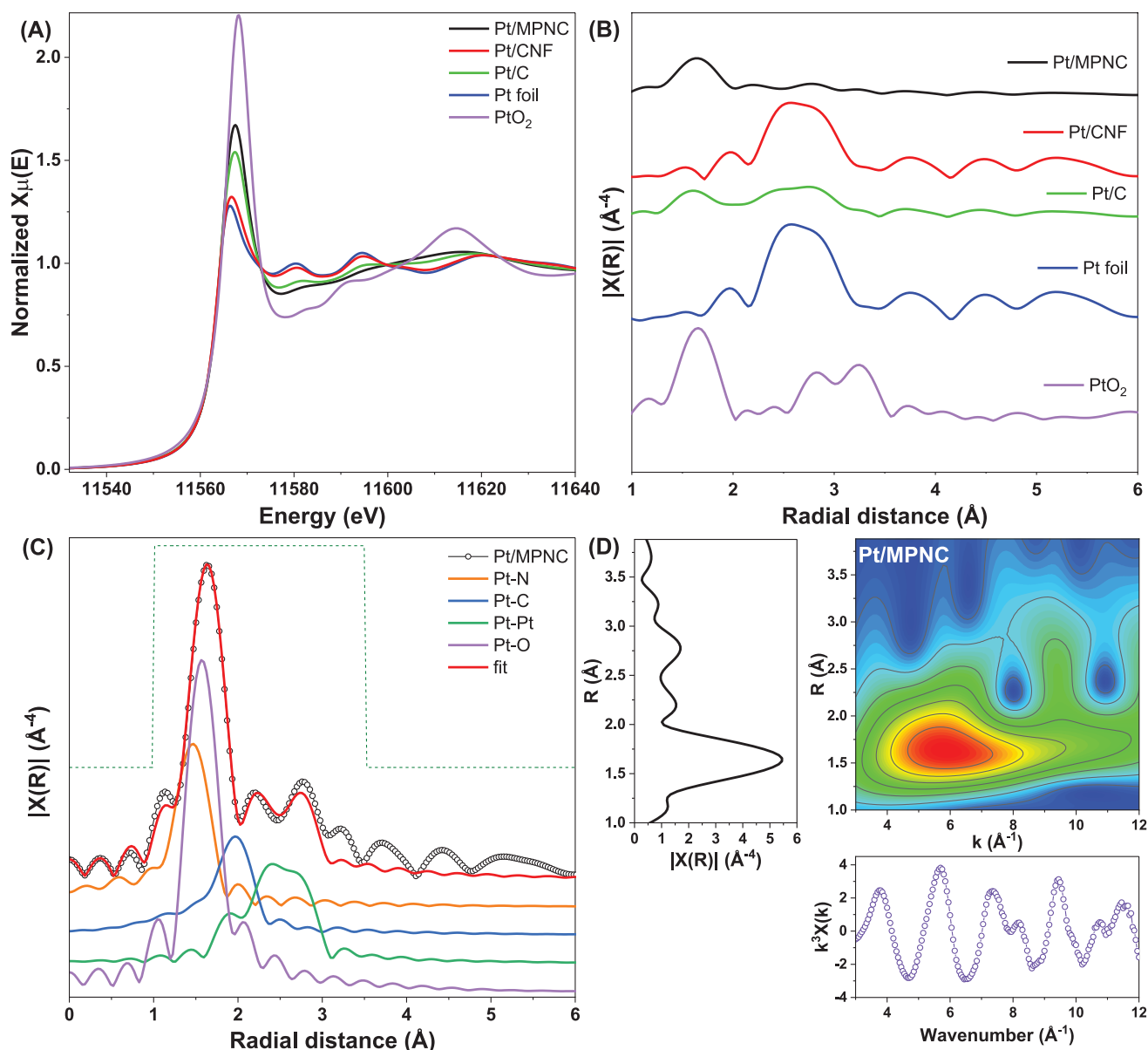


Figure 6. A) Pt L_{3} -edge XANES spectra of Pt/MPNC, Pt/CNF, Pt/C, Pt foil, and PtO_2 . B) Comparison of the Fourier-transform-EXAFS (FT-EXAFS) spectra of Pt/MPNC, Pt/CNF, Pt/C, Pt foil, and PtO_2 . C) Fitting of the FT-EXAFS spectra $\text{FT}[k^3\chi(k)]$ of Pt/MPNC with the fitting curve total signal (red line), and the experimental signal (black circles). The dashed green line represents the fitting range on the Å scale. D) Morlet wavelet transformation ($\eta = 5$ and $\sigma = 1$) of Pt L_{3} -edge EXAFS spectra of Pt/MPNC compared with its FT-EXAFS and k space spectra.

20 wt% loaded Pt/MPNC and Pt/CNF as well as of a commercial Pt/C (20 wt% Pt) reference catalyst in H_2 -saturated 0.5 M H_2SO_4 by a linear sweep voltammetry (LSV) technique using a three-electrode rotating disc electrode system, in which these three catalysts have the same Pt loading ($12.2 \mu\text{g}_{\text{Pt}} \text{ cm}^{-2}_{\text{geo}}$) on the electrode. The details of electrochemical measurements can be found in the experimental part.^[8]

From a practical perspective of water electrolysis, geometric current density, i.e., $\text{mA cm}^{-2}_{\text{geo}}$, and mass_{Pt} activity, i.e., $\text{A mg}^{-1}_{\text{Pt}}$, are important parameters. **Figure 7A** shows that Pt/MPNC exhibits the highest geometric current density under identical Pt loading on the electrode. And the geometric current density of Pt/MPNC ($68 \pm 4.2 \text{ mA cm}^{-2}_{\text{geo}}$) at the over-

potential of 20 mV vs. RHE is up to 4.2 and 26 times that of the commercial Pt/C ($16 \pm 2 \text{ mA cm}^{-2}_{\text{geo}}$) and the Pt/CNF ($2.6 \pm 0.9 \text{ mA cm}^{-2}_{\text{geo}}$) reference catalysts, respectively. The overpotentials of these three catalysts at $10 \text{ mA cm}^{-2}_{\text{geo}}$ were compared in **Figure 7B**, in which Pt/MPNC shows the lowest overpotential (only $12 \pm 0.5 \text{ mV}$ compared to $38.5 \pm 3.5 \text{ mV}$ for Pt/CNF) among these catalysts. In addition, Pt/MPNC also reveals the highest HER Pt mass activity of $5.45 \pm 0.35 \text{ A mg}^{-1}_{\text{Pt}}$ (at -20 mV vs. RHE) compared with the other two catalysts ($0.21 \pm 0.08 \text{ A mg}^{-1}_{\text{Pt}}$ for Pt/CNF and $1.28 \pm 0.16 \text{ A mg}^{-1}_{\text{Pt}}$ for Pt/C) in **Figure 7C,D**. Such improved HER Pt mass activity was assigned to the large ECSA ($\text{ECSA}_{\text{H}}: 80 \pm 6.9 \text{ m}^2 \text{ g}^{-1}_{\text{Pt}}$; $\text{ECSA}_{\text{CO}}: 136 \pm 5.3 \text{ m}^2 \text{ g}^{-1}_{\text{Pt}}$) of the Pt/MPNC catalyst (**Figure 4B**) and the

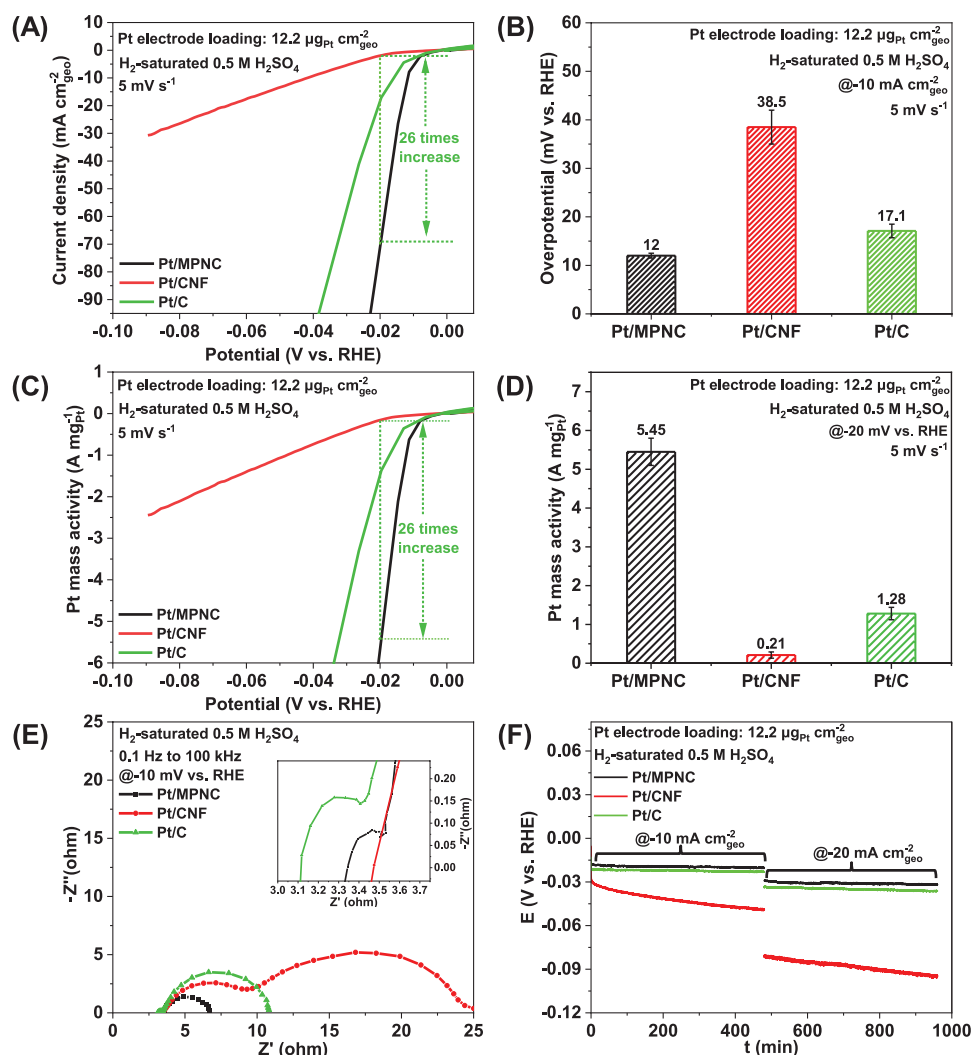


Figure 7. Electrochemical performance characterization of Pt/MPNC, Pt/CNF and Pt/C catalysts with the same Pt electrode loading ($12.2 \mu\text{g}_{\text{Pt}} \text{cm}^{-2}_{\text{geo}}$) in H_2 -saturated 0.5 M H_2SO_4 at a rotation speed of 1600 rpm, except for the chronopotentiometric curve (without rotation). A) Linear sweep voltammogram (LSV) curves normalized by the geometric surface area of the electrode, B) overpotentials at 10 $\text{mA cm}^{-2}_{\text{geo}}$, C) LSV curves normalized by used Pt mass, D) Pt mass activity at -20 mV vs. RHE, and E) potentiostatic electrochemical impedances of Pt/MPNC, Pt/CNF and Pt/C catalysts. F) Chronopotentiometric curve of Pt/MPNC, Pt/CNF, and Pt/C. The LSV curves were recorded at a scanning speed of 5 mV s^{-1} and corrected with 85% iR-compensation.

presence of a high density of Pt single atoms (Pt SA), which is demonstrated by the following analysis.

First, the ECSAs of the three catalysts were determined by H_{upd} and CO stripping and Pt/MPNC always shows the largest ECSA among these three catalysts as a result of the highly dispersed Pt entities formed on the MPNC support composed of Pt single atoms, clusters, and NPs, regardless of whether it is based on H_{upd} or CO stripping.

Additionally, a rough quantitative analysis of the contribution of Pt single atoms to the HER activity enhancement based on our dataset and various assumptions is provided in Supporting information S12. In this analysis, we estimate that the Pt single atoms of Pt/MPNC account for around 65% of the measured current, and that the Pt NPs and clusters of Pt/MPNC account for the remaining 35%. Hence, the improved geometric and Pt mass activity of Pt/MPNC can be ascribed to the large ECSA (to

be understood as total ECSA and not only apparent ECSA) and the presence of a high density of Pt single atoms.

In terms of HER mechanism, the HER process is thought to involve three elementary steps in acidic electrolytes, namely: i) Volmer step: $\text{H}_3\text{O}^+ + \text{e}^- \rightarrow \text{H}_{\text{ads}} + \text{H}_2\text{O}$, ii) Heyrovsky step: $\text{H}_{\text{ads}} + \text{H}_3\text{O}^+ + \text{e}^- \rightarrow \text{H}_2 + \text{H}_2\text{O}$, and iii) Tafel step: $\text{H}_{\text{ads}} + \text{H}_{\text{ads}} \rightarrow \text{H}_2$ and follows either the Volmer–Heyrovsky or the Volmer–Tafel pathway. Considering that many previous studies have shown that the Volmer–Tafel mechanism is preferred for Pt entities (including Pt NPs, Pt clusters, and Pt single atoms) in acidic electrolytes based on density functional theory considerations,^[7,71,72] it is reasonable to assume a similar situation in the case of this Pt/MPNC catalyst.

In order to demonstrate that the HER performance is based on the high ECSA and highly dispersed ultrasmall Pt entities formed on the MPNC support, and to exclude other factors,

first, the two N-doped carbon supports, MPNC and CNF without Pt, were demonstrated to be rather inactive for HER as shown in Figure S22 (Supporting Information), suggesting that the excellent HER catalytic activity is purely attributed to the Pt entities rather than to the carbon supports itself.

Second, considering that ECSA_H of Pt single atoms cannot be easily determined, the commonly used specific activity (calculated by the detected ECSA_H) is redefined to an apparent specific activity (ASA). Although ASA does not describe the entire catalyst specific activity, ASA was used to reflect the intrinsic HER activities of these three catalysts (Figure S23, Supporting Information). In that context, Pt/MPNC does not exhibit an obvious improved ASA compared with the other two catalysts, implying that the intrinsic activity of the Pt entities addressable in ECSA_H measurements is not the reason resulting in the increased geometric and Pt mass activity.

In addition, hydrophilicity of the catalyst powders and electrode thin films were investigated by water sorption and contact angle measurements, respectively.

Surprisingly, Pt/CNF shows a higher water absorption capacity (Figure S25, Supporting Information) than Pt/MPNC although an around 10 times smaller BET surface area and a similar surface composition of the pristine support (MPNC vs. CNF) was found (see Figure S16 in the Supporting Information). This could be explained on the one hand by the highly hydrophilic Pt,^[73] which is present on the outside of the CNF support and not mainly inside the porous network as in the MPNC support, and on the other hand by the different morphology of the supports. In the comparison of Pt/MPNC with Pt/C, the more hydrophilic character of the MPNC support leads to a stronger water uptake due to a higher surface functionalization (see Figure S16 in the Supporting Information).

After adding the ionomer solution to the catalyst powder, mixing the suspension (see Section 4; catalyst film preparation, same ionomer to catalyst mass ratio for all catalysts) and measuring the contact angle of the resulting catalyst films, a more hydrophilic nature of the Pt/CNF based electrode film ($74.3^\circ \pm 0.9^\circ$) is found compared to Pt/MPNC based electrode film ($131.4^\circ \pm 0.5^\circ$) as shown in Figure S26 (Supporting Information). In contrast, Pt/MPNC and Pt/C show comparable results ($130.3^\circ \pm 0.9^\circ$ for Pt/C and $131.4^\circ \pm 0.5^\circ$ for Pt/MPNC). It is worth noting that great attention must be paid to the interpretation of the absolute data obtained as hydrophilicity of Pt based carbon electrocatalysts in catalyst layers depends on multiple factors, such as morphology, surface composition, Pt loading and ionomer composition.^[74] Thus, solely the trend of the hydrophilicity in the order of Pt/CNF > Pt/MPNC ≈ Pt/C is discussed in regard to electrochemical HER performance in an aqueous electrolyte.

Recent reports have summarized that a hydrophilic surface features a high surface energy that attracts water^[75] and allows wetting of the surface. Furthermore, an aerophobic surface features rapid gas removal from the electrode layer to maximize the transport of reactants (i.e., protons) and products (i.e., H₂) and thus increase the electrocatalytic activity.^[76] It has been shown that an increased hydrophilicity as determined by contact angle measurements with constant active species causes an increased aerophobicity in HER.^[76] Therefore, the analysis of the hydrophilicity of the electrocatalysts by means of contact angle measurements and water sorption measurements is a further indicator of

the catalyst structure-activity correlation. Based solely on hydrophilicity considerations, a higher HER activity due to a better wetting should thus be found for Pt/CNF when compared to Pt/MPNC or Pt/C and equal activities for Pt/MPNC and Pt/C. However, this is not the case and this result only confirms that the improved HER activity obtained for the Pt/MPNC catalyst is not due to a better wettability but really relates to the ultrasmall and highly dispersed Pt species with high ECSA advantageously supported in highly porous MPNC nanospheres.

In addition to polarization curves, potentiostatic electrochemical impedance spectroscopy (PEIS) was also employed to disentangle the origin of the observed electrochemical activity trend. For these three catalysts, the PEIS experiments were conducted between 0.1 Hz and 100 kHz at an overpotential of 10 mV vs. RHE. Each Nyquist plot in Figure 7E exhibits two semicircles. The first semicircle at high frequency is overpotential-independent as shown in Figure S19 (Supporting Information) and could be related to the surface porous structure.^[77,78] The second one at low frequency is related to the charge transfer resistance during the HER process; the smaller this second semicircle is, the smaller the charge transfer resistance is. Pt/MPNC shows the lowest charge transfer resistance, suggesting that the least energy is used for electron transfer from Pt catalysts to protons from the electrolyte, resulting in the highest activity among these three catalysts. For Pt/CNF, regardless of the first or second one, it always shows the largest semicircles among these three catalysts, demonstrating that it has the poorest HER catalytic activity.

Finally, based on the numerous analysis regarding the structural and activity of the developed Pt/MPNC catalysts, and considering the negligible HER activity of the pristine MPNC supports, as well as the hydrophilicity and EIS analysis, we attribute the excellent HER performance of the Pt/MPNC catalyst to its highly dispersed Pt entities and advantageous nanostructure.

To investigate the effect of geometric Pt electrode loading, i.e., $\mu\text{g}_{\text{Pt}} \text{ per cm}^2_{\text{geo}}$, different Pt electrode loadings of Pt/MPNC were studied in H₂-saturated 0.5 M H₂SO₄ (see Figure 8A). We found that by decreasing the Pt electrode loading of Pt/MPNC from 577 to 0.9 $\mu\text{g}_{\text{Pt}} \text{ cm}^{-2}_{\text{geo}}$, the geometric activity decreases (see Figure S24 in the Supporting Information), while however, the mass activities normalized by both the mass of Pt and the total mass of catalysts (i.e., Pt and supporting material loadings) increases significantly specifically from 5.9 ± 0.9 to $56 \pm 3 \text{ A mg}^{-1}_{\text{Pt}}$ and from 1.18 ± 0.2 to $11.2 \pm 0.6 \text{ A mg}^{-1}_{\text{Cat}}$ at an overpotential of 50 mV vs. RHE (see Figure 8A), which we attribute to the H₂-blocked active sites. Pt is a very active catalyst for HER and the resulting H₂ during the HER process in the electrode can block the access to the HER active sites. When the Pt electrode loading decreases, the thickness of the catalyst layer on the electrode also decreases and H₂ is transported away within and from the rotating disc electrode (RDE) catalyst layers more quickly, therefore, more Pt active sites would be exposed and the Pt mass activity and catalyst mass activities would increase as the Pt electrode loading decreases.

By comparing the mass activities of Pt/MPNC with reported Pt single-atom catalysts at similar geometric Pt electrode loadings (under 4 $\mu\text{g}_{\text{Pt}} \text{ cm}^{-2}_{\text{geo}}$) shown in Figure 8B and Table S5 (Supporting Information), in which the mass activities at −50 mV vs. RHE are normalized by both the mass of Pt and the

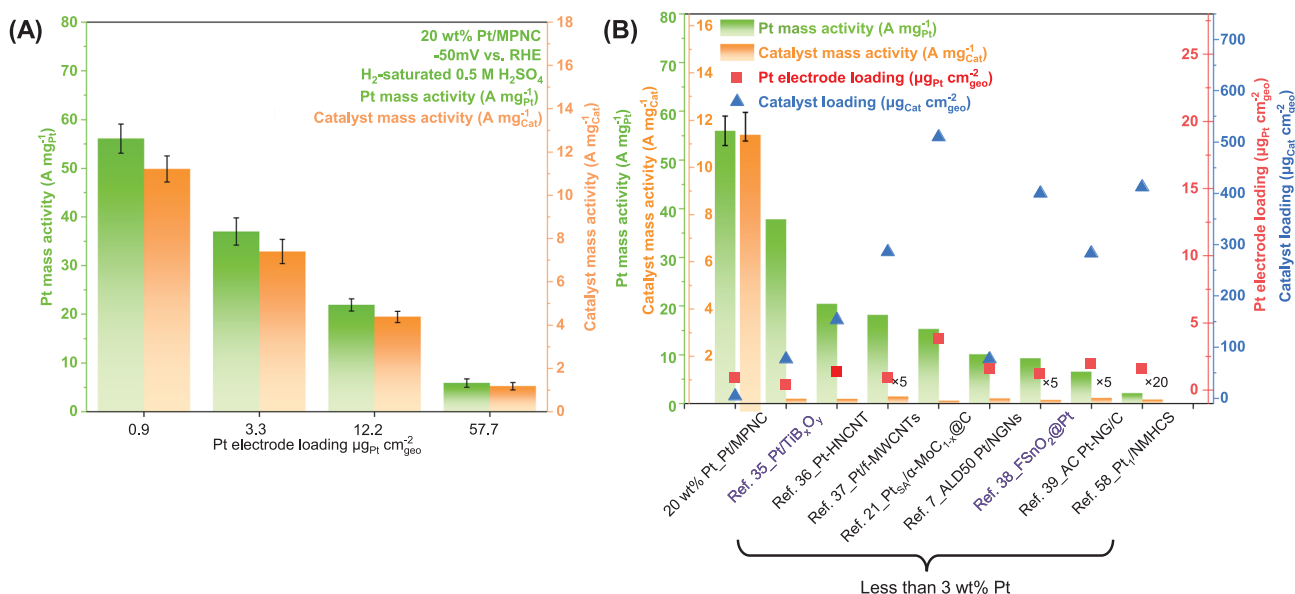


Figure 8. A) Pt and catalyst mass activities of Pt/MPNC for several Pt electrode loadings at -50 mV vs. RHE in H_2 -saturated 0.5 M H_2SO_4 . B) Mass activity comparison between Pt/MPNC with state-of-the-art single-atom HER electrocatalysts featuring similar Pt electrode loading on carbon-based supports (black) or non-carbon-based supports (violet) at -50 mV vs. RHE in H_2 -saturated 0.5 M H_2SO_4 . A more detailed parameter comparison can be found in Table S5 (Supporting Information).

total mass of catalyst, higher mass activities were found, which are respectively 1.5 ± 0.1 and 58 ± 3 times higher than the highest Pt mass activity and the highest catalyst mass activity of any Pt single-atom catalyst reported so far.^[35–38] This was unexpected as it is known that single atoms achieve maximum atom utilization,^[79] and that Pt single atoms are thought to be more active than Pt NPs and clusters,^[7,58,80] therefore, Pt single atoms should have higher Pt mass activity than Pt clusters/NPs which are present in the Pt/MPNC catalyst.

Since the HER process is always limited by mass transport for Pt catalysts in acidic electrolytes due to the fast kinetic process even at low overpotential, the thinner the catalyst film, the faster the mass transfer (H_2 gas and H^+ ion mass transfer), and the higher the Pt mass activity.^[8] In order to explain the improved Pt mass activity of Pt/MPNC, the Pt concentration, Pt electrode loading and corresponding catalyst loading were compared in Figure 8B and Table S5 (Supporting Information). The red squares represent the Pt electrode loading and the blue triangles represent the catalyst loading (i.e., combined Pt and support loadings). Although all catalysts have a similar Pt electrode loading (under $4 \mu\text{g}_{\text{Pt}} \text{cm}^{-2}_{\text{geo}}$), they show huge differences in catalyst loadings owing to various Pt concentrations. The single-atom catalysts have very low Pt concentrations (less than 3 wt%), in contrast to Pt/MPNC which has a higher Pt concentration of 20 wt% on the carbon support. Therefore, for similar Pt electrode mass loadings, the overall catalyst loadings for Pt single-atom catalysts on the electrode are much higher than for the Pt/MPNC catalyst, resulting in thicker catalyst films on the supporting electrodes and hence poor mass transport (both for H_2 gas and H^+ ion transport), thereby severely limiting the mass activity.

In addition to the excellent geometric and mass activity and low electrochemical resistance, Pt/MPNC also exhibited good stability as shown in Figure 7F where the chronopotentiometry

HER tests of these three catalysts were measured at two constant current densities (10 and $20 \text{ mA cm}^{-2}_{\text{geo}}$) for 16 h, respectively. Pt/MPNC and Pt/C exhibit an outstanding stability during the chronopotentiometry test, with a slight overpotential increase of only 2.6 mV vs. RHE after 8 h of chronopotentiometry test at $10 \text{ mA cm}^{-2}_{\text{geo}}$ and only 2.8 mV vs. RHE after 8 h of chronopotentiometry test at $20 \text{ mA cm}^{-2}_{\text{geo}}$. In contrast, the overpotential of Pt/CNF respectively increases by 17.5 and 14.6 mV vs. RHE after a 16 h chronopotentiometry test at 10 and $20 \text{ mA cm}^{-2}_{\text{geo}}$.

3. Conclusion

In summary, we synthesized highly dispersed Pt entities composed of Pt single atoms, clusters, and NPs on mesoporous N-doped carbon (MPNC) nanospheres by incipient wetness impregnation and subsequent Pt precursor reduction. We found that the mesopore-confinement effect and N-doping of MPNC nanospheres are favorable for the formation of highly dispersed Pt entities, as demonstrated by a combined approach of XRD, XPS, ac-ADF-STEM, XAS, and electrochemical CO stripping. These highly dispersed Pt entities endowed Pt/MPNC with an excellent HER geometric and Pt mass activity which is respectively 4.2 and 26 times higher than that of a commercial Pt/C and Pt/CNF catalyst at similar Pt loadings ($12.2 \mu\text{g}_{\text{Pt}} \text{cm}^{-2}_{\text{geo}}$). The improved HER activity is attributed to the morphological advantages of the MPNC support and the presence of Pt single atoms which were estimated to contribute to around 65% of the measured current, while the Pt NPs and clusters account for the remaining 35%. Surprisingly, by comparing different Pt electrode loadings of Pt/MPNC, we found that the geometric activity of Pt/MPNC decreases as the Pt electrode loading decreases, however, the Pt mass and catalyst mass

activities increase significantly. After optimization of the Pt electrode loading, Pt/MPNC exhibited ultrahigh mass activities ($56 \pm 3 \text{ A mg}^{-1}_{\text{Pt}}$ and $11.7 \pm 0.6 \text{ A mg}^{-1}_{\text{Cat}}$ at -50 mV vs. RHE), which are respectively 1.5 ± 0.1 and 58 ± 3 times higher than the highest Pt mass activity and the highest catalyst mass activity of Pt single-atom and cluster-based catalysts reported so far.

4. Experimental Section

Materials and Synthesis—Synthesis of MPNC Nanospheres: MPNC nanospheres were synthesized based on a modified protocol that was previously developed in our group.^[29,30] In brief, first, aniline (AppliChem, 99.5%, 28 mL) and SiO_2 NPs (Ludox TMA, Sigma-Aldrich, 23 nm, 34 wt% in H_2O , 33.7 g) were homogenized in 3150 mL of 0.11 M HCl solution (prepared by dilution from 5 M HCl, Alfa Aesar, with Milli-Q water, 18.2 MΩ) at 0°C . Second, ammonium persulfate (APS, Sigma-Aldrich, $\geq 98.0\%$, 70.1 g) was dissolved in 140 mL of 1 M HCl. The solution was added dropwise to the reaction mixture under vigorous stirring for 24 h in an ice bath at 0°C . After the completion of the reaction, we obtained a dark green solution that contains the suspension of the solid PANI/ SiO_2 composite. We subsequently separated the solid precipitate from the liquid phase by centrifugation followed by washing with Milli-Q water until the supernatant reached a neutral pH value. The wet phase made of the composite material was then freeze-dried after which it was subjected to carbonization in a tube furnace (Carbolite Gero GmbH & Co. KG) under a constant N_2 gas flow of 120 mL min^{-1} and at a controlled temperature of 1000°C (the furnace was sat to reach the target temperature by setting the ramp at 200°C h^{-1} , and dwell time = 6 h). The resulting MPNC/ SiO_2 material was dispersed in a 4 M ammonium hydrogen difluoride (NH_4HF_2 , Carl Roth, crystalline product dissolved in Milli-Q water) solution and stirred for 48 h at room temperature. For every 1 g of MPNC/ SiO_2 , 50 mL of the etching solution made of ammonium hydrogen difluoride was used. The solution was then removed by filtration and the residual material was washed with Milli-Q water. Then, the product was taken up in ethanol (abs., VWR) and stirred for 24 h in order to dilute residual water in the porous system by diffusion and to ensure mild drying conditions in air after removal of ethanol from the carbon. The synthesis yielded 12.5 g of final MPNC material.

Materials and Synthesis—Synthesis of CNF: CNF was prepared by an oxidative polymerization of aniline in an acetic acid solution followed by a carbonization process. In the first vessel which was kept at 0°C , we dissolved 8 mL aniline solution into 900 mL of 0.4 M acetic acid solution, which was prepared by dilution from 100% acetic acid stock solution (VWR Chemicals). In the second step, a solution mixture made of 20 g APS in 50 mL of 0.4 M acetic acid was added dropwise to the first vessel reaction mixture within 15 min under constant stirring at 170 rpm. The reaction mixture was continuously stirred for 24 h in an ice bath at 0°C . Thereafter, it was possible to visualize the formation of the solid phase. Subsequently, the crude product was separated from the reaction solution by centrifugation and washed with Milli-Q water until the supernatant reached neutral pH. Upon recovery, the dark green solid phase was then freeze-dried for four days. The obtained CNF was then carbonized under N_2 flow (120 mL min^{-1}) at 1000°C in a tubular furnace (800P, Linn High Therm GmbH, one side open tube). Note that we used the same parameters implemented during the carbonization of the MPNC, i.e., heating rate and dwell time. By completing the synthesis, we managed to make 2.5 g of CNF, which we describe as a reasonable yield.

Materials and Synthesis—Synthesis of Pt/MPNC and Pt/CNF: Pt/MPNC and Pt/CNF were synthesized by an incipient wetness impregnation process and a thermal reduction. A homemade fluidized bed reactor was used for a thermal reduction process as depicted in Figure S1 (Supporting Information), in which more homogeneous Pt NPs distribution was achieved in comparison to previous studies that used a conventional tube furnace,^[19] which was attributed to the uniform flow of the reducing gas (10 vol% H_2/Ar). First, the MPNC and CNF support materials were freeze-dried for 10 h prior to being used as a support

for the Pt electrocatalysts. Hereafter, we made a solution by dissolving 100 mg $\text{H}_2\text{PtCl}_6 \cdot 6\text{H}_2\text{O}$ (99.95%, Alfa Aesar) into 1054 μL of 1:1 v:v solution of isopropanol and Milli-Q water. In the next step, we added the content of the previous solution in a dropwise fashion into a flask that contains 150 mg of MPNC to make Pt/MPNC or CNF to make Pt/CNF. Subsequently, the Pt impregnated carbons were treated in an ultrasonic bath at 0°C for 30 min to ensure optimal Pt distribution. Consecutively, the solvents were removed by freeze-drying for 24 h. Pt/MPNC and Pt/CNF were yielded by reduction of impregnated Pt precursor in 10 vol% H_2/Ar at 300°C for 2 h (ramp 600°C h^{-1}). The Pt loadings on Pt/MPNC and Pt/CNF were determined by thermogravimetric analysis (TGA) as shown in Figure S16B (Supporting Information).

Physical and Chemical Characterization—Structure Characterization: Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), and Scanning Transmission Electron Microscopy (STEM): Field emission SEM was used to obtain low magnification images of Pt and the support. SEM measurements were conducted using an acceleration voltage of 5 kV at a working distance of $\approx 3 \text{ mm}$ with an FEG-HRSEM SU8220 (Hitachi). HR-TEM measurements were accomplished using a Talos 200X electron microscope (Thermo Fisher) operated at an acceleration voltage of 200 kV. For SEM and TEM measurements, all samples were loaded onto a standard holey carbon film-coated on a Cu metallic grid (300 mesh, Plano GmbH). The structure, size, dispersity, and morphology of Pt entities were investigated using STEM in a double aberration-corrected (ac) JEOL JEM-ARM200F TEM/STEM, operating at 200 kV equipped with a Gatan Quantum EELS. Annular dark-field ac-(ADF)-STEM and bright-field ac-(BF)-STEM images were recorded simultaneously with a probe current of 23 pA and a convergence semiangle of $\approx 16 \text{ mrad}$. The inner collection semiangle for ADF-STEM imaging was 38 mrad.

Physical and Chemical Characterization— N_2 Physisorption Measurements: Surface area, pore volume, and pore size distribution were determined from N_2 physisorption isotherms obtained at 77 K with a 3P 300 Micro using QSDFT model taking into account spherical, cylindrical, and split pores. All measurements were done by degassing MPNC and CNF samples at 200°C for 4 h under high vacuum.

Physical and Chemical Characterization—Water Sorption Measurements: H_2O sorption measurements were performed using a Vstar4 analyzer from Quantachrome. Prior to measurement, the samples were degassed for 16 h under a high vacuum at 150°C . The isotherms were measured at 25°C for a relative pressure P/P_0 ramped from 0.01 to 0.95 for the adsorption branch and from 0.95 to 0.1 for the desorption branch.

Physical and Chemical Characterization—Elemental Analysis: Elemental analysis (EA) of the following elements C, N, H, and S was conducted using a Vario MICRO Cube System (Elementar Analysensysteme GmbH, Germany) by combusting samples at 1150°C in O_2 atmosphere and quantitative detection of the combustion gases (N_2 , H_2O , CO_2 , and SO_2).

Physical and Chemical Characterization—Raman Spectroscopy Measurements: All Raman spectra were recorded using a Senterra II Raman microscope (Bruker Optics GmbH, Ettlingen, Germany) equipped with a 532 nm laser (0.25 mW). Measurements were done with a wavenumber resolution of 4 cm^{-1} . Each point was tested with 500 scans where each spectrum integration time was set for 10 s and the final Raman spectrum was the sum of 500 integrated spectra.

Physical and Chemical Characterization—X-Ray Photoelectron Spectroscopy: X-ray photoelectron spectroscopy (XPS) was measured using a SPECS EnviroESCA with the detector of 1D delay line detector (1D-DLD). Samples were placed with an angle of 55° in the beam. Source energy of 1486.6 eV was used for the XPS measurement. The shifts in binding energy in the samples caused by possible uncompensated charging effects were compensated for by normalizing to 284.8 eV (for the position of sp^2 (C—C) of the C 1s signal). Data were analyzed using the XPSpeak 41.

Physical and Chemical Characterization—X-Ray Diffraction and Rietveld Refinement: XRD characterization of Pt loaded carbon catalysts was performed in rotating glass capillaries (inner diameter 0.5 mm, Hilgenberg) on a STOE Stadi P diffractometer with Cu $\text{K}\alpha 1$ radiation ($\lambda = 0.154060 \text{ nm}$), a Ge (111) monochromator, and a linear position

sensitive detector (PSD). The X-ray gun was operated at 40 kV and 30 mA. The samples were measured in Debye-Scherrer geometry from 30° to 105° (2 θ) using a step width of 0.5° and a step time of 120 s. Rietveld refinements were conducted on FullProf software (Version 6.30—Sep2018-ILL JRC).^[44]

Physical and Chemical Characterization—X-Ray Absorption Spectroscopy: X-ray absorption spectroscopy (XAS) data of Pt foil Pt/MPNC, Pt/C, Pt/CNF, and PtO₂ were collected at the SuperXAS beamline at the Paul Scherrer Institute (PSI), which operates in top-up mode at 2.4 GeV and a ring current of 400 mA. The polychromatic X-rays from a 2.9 T superbend magnet were collimated by a silicon coated mirror (which also served to reduce higher order harmonics) and subsequently monochromatized by a Si (111) channel-cut crystal. The monochromator was operated at a frequency of 1 Hz resulting in two spectra per second. The beam was focused by an Rh-coated toroidal mirror. The Pt L₃-edge X-ray absorption spectra were collected in transmission mode using 15 cm long ionization chambers filled with 2 bar of N₂ and by measuring a Pt foil simultaneously for absolute energy calibration.

Background subtraction, interpolation and averaging were done with the python based software ProQEXAFS.^[81]

The measured k^3 -weighted extended X-ray absorption fine structure (EXAFS) spectra $k^3\chi(k)$ were extracted after normalization in ATHENA.^[82] The spectra were reduced into the range $\Delta k \approx 3$ –12 Å^{−1} and Fourier transformed to $FT[k^3\chi(k)]$ into the real-space interval $\Delta R \approx 0$ –6 Å.

Physical and Chemical Characterization—Contact Angle Measurements: Contact angle measurement was performed using a DataPhysics optical contact angle measuring and contour analysis system. First, the electrocatalyst ink was prepared by dispersing 3 mg of the catalysts in a mixed solvent made by mixing 1.99 mL Milli-Q water (18.2 MΩ) with 0.5 mL high-purity isopropanol (99.9%, Sigma Aldrich) and 10 μL Nafion solution (5 wt%, Sigma Aldrich). Then, the electrocatalyst ink (10 μL) was pipetted onto a glassy carbon (GC) rotating disc electrode (RDE) and then rotated at a speed of 400 rpm at ambient temperature to form a homogeneous film. Subsequently, the electrode was placed in an oven at 60 °C for 10 min. Then, a droplet of 10 μL milli-Q water was placed on the electrocatalyst film for optical contact angle measurement.

Physical and Chemical Characterization—Electrochemical Performance Measurements: All electrochemical experiments were recorded using a Biologic SP200 potentiostat. The HER electrocatalytic activity was evaluated in a temperature-controlled three-electrode cell set-up where a RDE acted as the working electrode, whereas a reversible hydrogen electrode (RHE) from Gaskatel (Germany) and a Pt mesh (12 cm²) were used as reference and counter electrodes, respectively. The electrocatalyst ink was prepared by dispersing 3 mg of the catalysts in a mixed solvent made by mixing 1.99 mL Milli-Q water (18.2 MΩ) with 0.5 mL high-purity isopropanol (99.9%, Sigma Aldrich) and 10 μL Nafion solution (5 wt%, Sigma Aldrich) in which the Nafion solution mainly acts as a binder. Resulting in a mass ratio of ionomer to carbon and ionomer to Pt of 0.18 and 0.73, respectively. The solution was then homogenized by ultrasonication for 10 min in an ice bath. In the next stage, we pipetted 10 μL of the catalyst ink, if not otherwise stated, onto the prepolished GC RDE with an area of 0.196 cm² leading to a Pt loading of 12.2 μg

Pt cm^{−2}_{geo}. To achieve homogeneous deposition on the GC surface, the RDE was first positioned upside down then the ink was applied to the GC surface, which was then rotated at a speed of 400 rpm at ambient temperature. Subsequently, the electrode was placed in an oven (Mettmert GmbH) at 60 °C for 10 min. The Pt-based catalysts were conditioned before activity measurements by running 50 cycles of cyclic voltammetry (CV) between 0.05 and 1 V vs. RHE with a scan rate of 100 mV s^{−1} in 0.5 M H₂SO₄ electrolyte saturated with ultrahigh purity N₂ (≥99.999%). Then, CV traces from 0.05 to 1 V vs. RHE at 20 mV s^{−1} under N₂ saturation were collected to compare hydrogen underpotential deposition features. To examine the performance of the Pt-based electrocatalysts for HER, polarization curves were recorded at a rotation speed of 1600 rpm and a scanning rate of 5 mV s^{−1} in a 0.5 M H₂SO₄ electrolyte, which was saturated with ultrahigh purity H₂ (≥99.999%, Sauerstoffwerk Friedrichshafen GmbH). The temperature was held constant at 25 °C during the course of the experiment and

the electrolyte resistance was corrected with 85% iR-compensation as determined by an electrochemical impedance measurement at a high frequency of 100 kHz.

A nonrotated temperature-controlled three-electrode cell setup was used to evaluate catalyst stability by chronopotentiometry measurements at two constant current densities (−10 and −20 mA cm^{−2}_{geo}) for 8 h in H₂-saturated 0.5 M H₂SO₄ electrolyte. Noteworthy, the catalyst film was deposited onto a flat GC plate, instead of a RDE electrode. The electrocatalyst ink was prepared by the same method as described above, then dropped on a flat GC plate (1 cm²) and dried naturally at ambient temperature for 2 h. Subsequently, the electrode was placed in an oven (Mettmert GmbH) at 60 °C for 10 min.

CO stripping experiments were conducted in CO-saturated 0.1 M HClO₄ electrolyte based on the following protocol. First, the electrolyte (0.1 M HClO₄) was saturated by purging CO for 7 min. Then the working electrode was inserted in the electrolyte and held at a potential of 0.1 V vs. RHE for 12 min. For the first 2 min, CO was purged through the electrolyte to adsorb on the electrode, followed by 10 min of N₂ purging, to remove CO from the electrolyte. After that, CV was started from 0.05 to 1.2 V vs. RHE with a scan rate of 20 mV s^{−1}.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

Z.Z. and S.K. contributed equally to this work. Financial support was provided by the German Federal Ministry of Education and Research (BMBF, FKZ: 03SF0614A (CORAL-HD), grantee A.F.), the Eva Mayr-Stihl Foundation within the Saltus!-program of the University of Freiburg (Saltus!-Group “Design of interfaces and interphases in energy materials”, University of Freiburg, grantee A.F.) and the Volkswagen Foundation (Momentum - Funding for Recently Tenured Professors, “In-Situ Electrochemical Transmission Electron Microscopy - Understanding Electrocatalysts Dynamics under Electrochemical Conditions at the Nano Scale”, grantee A.F.). Z.Z. also appreciates the Chinese Scholarship Council and the state-funded scholarships of the Landesgraduiertenförderung of Baden Württemberg. Many thanks to Dr. Bizan N. Balzer for his help in contact angle measurements and Dr. Yi Thomann and Dr. Ralf Thomann from the Core Facility “Imaging of Materials Systems” @ FIT for access to the electron microscopes and help with part of the STEM-EDS measurements. The authors acknowledge the Swiss Light Source (SLS) for providing beamtime at the SuperXAS beamline (Mesquick proposal 20222077). S.P. acknowledges NCCR Catalysis (grant number 180544), a National Centre of Competence in Research funded by the Swiss National Science Foundation for his funding. H.E.M.H. is grateful for the support provided by Electron Microscopy Research Technology Platform, Department of Chemistry (Prof. Julie Macpherson), and Department of Physics (Prof. Richard Beanland), University of Warwick, to access the transmission electron microscopy suite.

Open access funding enabled and organized by Projekt DEAL.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

electrochemical CO stripping, hydrogen evolution reaction, mesoporous N-doped carbon nanospheres, Pt clusters, Pt nanoparticles, Pt single atoms, ultrahigh mass activity

Received: September 25, 2022

Revised: January 28, 2023

Published online: March 22, 2023

- [1] J. Turner, G. Sverdrup, M. K. Mann, P. Maness, B. Kroposki, M. Ghirardi, R. J. Evans, D. Blake, *Int. J. Energy Res.* **2008**, 32, 379.
- [2] S. E. Hosseini, M. A. Wahid, *Renew Sustain Energy Rev* **2016**, 57, 850.
- [3] S. Su, Q. Zhou, Z. Zeng, D. Hu, X. Wang, M. Jin, X. Gao, R. Nötzel, G. Zhou, Z. Zhang, *ACS Appl. Mater. Interfaces* **2018**, 10, 8026.
- [4] S. S. Kumar, H. Lim, *Energy Rep.* **2022**, 8, 13793.
- [5] M. A. Hubert, L. A. King, T. F. Jaramillo, *ACS Energy Lett.* **2022**, 7, 17.
- [6] M. Chatenet, B. G. Pollet, D. R. Dekel, F. Dionigi, J. Deseure, P. Millet, R. D. Braatz, M. Z. Bazant, M. Eikerling, I. Staffell, P. Balcombe, Y. Shao-Horn, H. Schäfer, *Chem. Soc. Rev.* **2022**, 51, 4583.
- [7] N. Cheng, S. Stambula, D. Wang, M. N. Banis, J. Liu, A. Riese, B. Xiao, R. Li, T.-K. Sham, L.-M. Liu, G. A. Botton, X. Sun, *Nat. Commun.* **2016**, 7, 13638.
- [8] J. N. Hansen, H. Prats, K. K. Toudahl, N. Mørch Secher, K. Chan, J. Kibsgaard, I. Chorkendorff, *ACS Energy Lett.* **2021**, 6, 1175.
- [9] B. Pivovar, *Nat. Catal.* **2019**, 2, 562.
- [10] J. A. Dallas, S. Raval, S. Saydam, A. G. Dempster, *Acta Astronaut.* **2021**, 186, 74.
- [11] L. Zhang, K. Doyle-Davis, X. Sun, *Energy Environ. Sci.* **2019**, 12, 492.
- [12] S.-F. Zheng, J.-S. Hu, L.-S. Zhong, L.-J. Wan, W.-G. Song, *J. Phys. Chem. C* **2007**, 111, 11174.
- [13] Q. Xu, M. Chandra, *J. Power Sources* **2006**, 163, 364.
- [14] Z. W. Seh, J. Kibsgaard, C. F. Dickens, I. Chorkendorff, J. K. Nørskov, T. F. Jaramillo, *Science* **2017**, 355, eaad4998.
- [15] X. K. Wan, H. Bin Wu, B. Y. Guan, D. Luan, X. W. Lou, *Adv. Mater.* **2020**, 32, 1901349.
- [16] H. Zhang, P. An, W. Zhou, B. Y. Guan, P. Zhang, J. Dong, X. W. Lou, *Sci. Adv.* **2018**, 4, eaao6657.
- [17] F. Wu, Y. Xing, X. Zeng, Y. Yuan, X. Zhang, R. Shahbazian-Yassar, J. Wen, D. J. Miller, L. Li, R. Chen, J. Lu, K. Amine, *Adv. Funct. Mater.* **2016**, 26, 7626.
- [18] S. T. Christensen, J. W. Elam, F. A. Rabuffetti, Q. Ma, S. J. Weigand, B. Lee, S. Seifert, P. C. Stair, K. R. Poeppelmeier, M. C. Hersam, M. J. Bedzyk, *Small* **2009**, 5, 750.
- [19] R. Xu, Z. Zeng, Y. Lei, *Nat. Commun.* **2022**, 13, 2435.
- [20] Z. Zeng, R. Xu, H. Zhao, H. Zhang, L. Liu, S. Xu, Y. Lei, *Mater. Today Nano* **2018**, 3, 54.
- [21] W. Wang, Y. Wu, Y. Lin, J. Yao, X. Wu, C. Wu, X. Zuo, Q. Yang, B. Ge, L. Yang, G. Li, S. Chou, W. Li, Y. Jiang, *Adv. Funct. Mater.* **2022**, 32, 2108464.
- [22] C. Tsounis, B. Subhash, P. V. Kumar, N. M. Bedford, Y. Zhao, J. Shenoy, Z. Ma, D. Zhang, C. Y. Toe, S. Cheong, R. D. Tilley, X. Lu, L. Dai, Z. Han, R. Amal, *Adv. Funct. Mater.* **2022**, 2203067, 2203067.
- [23] L. Wen, R. Xu, C. Cui, W. Tang, Y. Mi, X. Lu, Z. Zeng, S. L. Suib, P.-X. Gao, Y. Lei, *Nano Lett.* **2018**, 18, 4914.
- [24] S. M. George, *Chem. Rev.* **2010**, 110, 111.
- [25] R. Tietema, *Comprehensive Mater. Process.* **2014**, 4, 519.
- [26] D. M. Mattox, *Met. Finish.* **2002**, 100, 394.
- [27] J. Regalbuto, *Catalyst Preparation: Science and Engineering*, CRC Press, Boca Raton, FL **2016**.
- [28] B. A. T. Mehrabadi, S. Eskandari, U. Khan, R. D. White, J. R. Regalbuto, *Adv. Catal.* **2017**, 61, 1.
- [29] J. Melke, J. Martin, M. Bruns, P. Hügenell, A. Schökel, S. M. Isaza, F. Fink, P. Elsässer, A. Fischer, *ACS Appl. Energy Mater.* **2020**, 3, 11627.
- [30] J. Melke, R. Schuster, S. Möbus, T. Jurzinsky, P. Elsässer, A. Heilemann, A. Fischer, *Carbon* **2019**, 146, 44.
- [31] G. Bae, D. H. Youn, S. Han, J. S. Lee, *Carbon* **2013**, 51, 274.
- [32] M. N. Groves, A. S. W. Chan, C. Malardier-Jugroot, M. Jugroot, *Chem. Phys. Lett.* **2009**, 481, 214.
- [33] Q. Guo, X. Lu, G. Fei, Z. Wang, H. Xia, *Ind. Eng. Chem. Res.* **2022**, 61, 1398.
- [34] Y. Lou, J. Xu, Y. Zhang, C. Pan, Y. Dong, Y. Zhu, *Mater. Today Nano* **2020**, 12, 100093.
- [35] X. Cheng, B. Xiao, Y. Chen, Y. Wang, L. Zheng, Y. Lu, H. Li, G. Chen, *ACS Catal.* **2022**, 12, 5970.
- [36] L. Zhang, Q. Wang, R. Si, Z. Song, X. Lin, M. N. Banis, K. Adair, J. Li, K. Doyle-Davis, R. Li, L. Liu, M. Gu, X. Sun, *Small* **2021**, 17, 2004453.
- [37] J. Ji, Y. Zhang, L. Tang, C. Liu, X. Gao, M. Sun, J. Zheng, M. Ling, C. Liang, Z. Lin, *Nano Energy* **2019**, 63, 103849.
- [38] T. Kim, S. B. Roy, S. Moon, S. Yoo, H. Choi, V. G. Parale, Y. Kim, J. Lee, S. C. Jun, K. Kang, S.-H. Chun, K. Kanamori, H.-H. Park, *ACS Nano* **2022**, 16, 1625.
- [39] M. Sun, J. Ji, M. Hu, M. Weng, Y. Zhang, H. Yu, J. Tang, J. Zheng, Z. Jiang, F. Pan, C. Liang, Z. Lin, *ACS Catal.* **2019**, 9, 8213.
- [40] D. Mattia, M. P. Rossi, B. M. Kim, G. Korneva, H. H. Bau, Y. Gogotsi, *J. Phys. Chem. B* **2006**, 110, 9850.
- [41] S. Gayathri, P. Arunkumar, E. J. Kim, S. Kim, I. Kang, J. H. Han, *Electrochim. Acta* **2019**, 318, 290.
- [42] O. L. Li, S. Chiba, Y. Wada, G. Panomsuwan, T. Ishizaki, *J. Mater. Chem. A* **2017**, 5, 2073.
- [43] A. Sadezky, H. Muckenhuber, H. Grothe, R. Niessner, U. Pöschl, *Carbon* **2005**, 43, 1731.
- [44] J. Martin, J. Melke, C. Njel, A. Schökel, J. Büttner, A. Fischer, *ChemElectroChem* **2021**, 8, 4835.
- [45] S. Mourdikoudis, R. M. Pallares, N. T. K. Thanh, *Nanoscale* **2018**, 10, 12871.
- [46] C. Tsounis, B. Subhash, P. V. Kumar, N. M. Bedford, Y. Zhao, J. Shenoy, Z. Ma, D. Zhang, C. Y. Toe, S. Cheong, R. D. Tilley, X. Lu, L. Dai, Z. Han, R. Amal, *Adv. Funct. Mater.* **2022**, 32, 2203067.
- [47] P. Kuang, Y. Wang, B. Zhu, F. Xia, C. W. Tung, J. Wu, H. M. Chen, J. Yu, *Adv. Mater.* **2021**, 33, 2008599.
- [48] E. G. Ciapina, S. F. Santos, E. R. Gonzalez, *J. Electroanal. Chem.* **2018**, 815, 47.
- [49] G. A. Kamat, J. A. Zamora Zeledón, G. T. K. K. Gunasooriya, S. M. Dull, J. T. Perryman, J. K. Nørskov, M. B. Stevens, T. F. Jaramillo, *Commun Chem* **2022**, 5, 20.
- [50] F. Maillard, S. Schreier, M. Hanzlik, E. R. Savinova, S. Weinkauff, U. Stimming, *Phys. Chem. Chem. Phys.* **2005**, 7, 375.
- [51] J.-F. Huang, R.-H. Zeng, J.-L. Chen, *J. Mater. Chem. A* **2021**, 9, 21972.
- [52] T. Biegler, D. A. J. Rand, R. Woods, *J. Electroanal. Chem. Interfacial Electrochem.* **1971**, 29, 269.
- [53] I. J. McPherson, P. A. Ash, L. Jones, A. Varambhia, R. M. J. Jacobs, K. A. Vincent, *J. Phys. Chem. C* **2017**, 121, 17176.
- [54] S. Taylor, E. Fabbri, P. Levecque, T. J. Schmidt, O. Conrad, *Electrocatalysis* **2016**, 7, 287.
- [55] M. Shao, J. H. Odell, S.-I. Choi, Y. Xia, *Electrochem. Commun.* **2013**, 31, 46.
- [56] F. Maillard, M. Eikerling, O. V. Cherstiouk, S. Schreier, E. Savinova, U. Stimming, *Faraday Discuss.* **2004**, 125, 357.
- [57] X. H. Li, M. Antonietti, *Chem. Soc. Rev.* **2013**, 42, 6593.
- [58] P. Kuang, Y. Wang, B. Zhu, F. Xia, C. Tung, J. Wu, H. M. Chen, J. Yu, *Adv. Mater.* **2021**, 33, 2008599.
- [59] J. Liu, M. Jiao, L. Lu, H. M. Barkholtz, Y. Li, Y. Wang, L. Jiang, Z. Wu, D. Liu, L. Zhuang, C. Ma, J. Zeng, B. Zhang, D. Su, P. Song, W. Xing, W. Xu, Y. Wang, Z. Jiang, G. Sun, *Nat. Commun.* **2017**, 8, 15938.

- [60] J. Melke, B. Peter, A. Habereeder, J. Ziegler, C. Fasel, A. Nefedov, H. Sezen, C. Wöll, H. Ehrenberg, C. Roth, *ACS Appl. Mater. Interfaces* **2016**, 8, 82.
- [61] X. Ning, Y. Li, B. Dong, H. Wang, H. Yu, F. Peng, Y. Yang, *J. Catal.* **2017**, 348, 100.
- [62] Y. Peng, B. Lu, N. Wang, L. Li, S. Chen, *Phys. Chem. Chem. Phys.* **2017**, 19, 9336.
- [63] J. Ma, A. Habrioux, Y. Luo, G. Ramos-Sanchez, L. Calvillo, G. Granozzi, P. B. Balbuena, N. Alonso-Vante, *J. Mater. Chem. A* **2015**, 3, 11891.
- [64] B. Qiao, A. Wang, X. Yang, L. F. Allard, Z. Jiang, Y. Cui, J. Liu, J. Li, T. Zhang, *Nat. Chem.* **2011**, 3, 634.
- [65] S. Yang, J. Kim, Y. J. Tak, A. Soon, H. Lee, *Angew. Chem., Int. Ed.* **2016**, 55, 2058.
- [66] H. Wei, X. Liu, A. Wang, L. Zhang, B. Qiao, X. Yang, Y. Huang, S. Miao, J. Liu, T. Zhang, *Nat. Commun.* **2014**, 5, 5634.
- [67] Y. Chen, P. Wang, H. Hao, J. Hong, H. Li, S. Ji, A. Li, R. Gao, J. Dong, X. Han, M. Liang, D. Wang, Y. Li, *J. Am. Chem. Soc.* **2021**, 143, 18643.
- [68] Z. Abdi, S. E. Balaghi, A. S. Sologubenko, M. G. Willinger, M. Vandichel, J. R. Shen, S. I. Allakhverdiev, G. R. Patzke, M. M. Najafpour, *ACS Sustainable Chem. Eng.* **2021**, 9, 9494.
- [69] V. L. Sushkevich, O. V. Safonova, D. Palagin, M. A. Newton, J. A. van Bokhoven, *Chem. Sci.* **2020**, 11, 5299.
- [70] H. Funke, A. C. Scheinost, M. Chukalina, *Phys. Rev. B* **2005**, 71, 094110.
- [71] G.-F. Wei, Z.-P. Liu, *Chem. Sci.* **2015**, 6, 1485.
- [72] D. T. Limmer, A. P. Willard, P. Madden, D. Chandler, *Proc. Natl. Acad. Sci. U. S. A.* **2013**, 110, 4200.
- [73] K. W. Bewig, W. A. Zisman, *J. Phys. Chem.* **1965**, 69, 4238.
- [74] W. Olbrich, T. Kadyk, U. Sauter, M. Eikerling, *J. Electrochem. Soc.* **2022**, 169, 054521.
- [75] G. Zhao, Z. Schwartz, M. Wieland, F. Rupp, J. Geis-Gerstorfer, D. L. Cochran, B. D. Boyan, *J. Biomed. Mater. Res., Part A* **2005**, 74A, 49.
- [76] B. K. Kim, M. J. Kim, J. J. Kim, *ACS Appl. Mater. Interfaces* **2021**, 13, 11940.
- [77] W. F. Chen, C. H. Wang, K. Sasaki, N. Marinkovic, W. Xu, J. T. Muckerman, Y. Zhu, R. R. Adzic, *Energy Environ. Sci.* **2013**, 6, 943.
- [78] B. Losiewicz, *Int. J. Hydrogen Energy* **2004**, 29, 145.
- [79] Y. Chen, S. Ji, C. Chen, Q. Peng, D. Wang, Y. Li, *Joule* **2018**, 2, 1242.
- [80] H. Luo, Y. Liu, S. D. Dimitrov, L. Steier, S. Guo, X. Li, J. Feng, F. Xie, Y. Fang, A. Sapelkin, X. Wang, M.-M. Titirici, *J. Mater. Chem. A* **2020**, 8, 14690.
- [81] A. H. Clark, J. Imbao, R. Frahm, M. Nachtegaal, *J. Synchrotron Radiat.* **2020**, 27, 551.
- [82] B. Ravel, M. Newville, *J. Synchrotron Radiat.* **2005**, 12, 537.